Filling the Gaps?: Private Sector Response to the Push and Pull of the Indian Public Schooling System

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October 29, 2026

Abstract

We study how the quality of public schools, measured by the average test score outcomes they produce, shapes the evolution of local schooling markets in rural India. Combining district-year ASER learning levels for public-school children with a reconstructed panel of the universe of schools in UDISE for 2012-2017, we document a robust inverted-U relationship between lagged public school quality and the size of the private sector. Where public schools perform poorly, marginal improvements in public learning are associated with more private school entry and a larger stock of private schools; beyond a turning point located near the tenth to twenty-fifth percentile of the score distribution, the relationship reverses, so that for district-years with better public learning, the stock of private schools and private school entry decline as public quality improves. The shape is not a parametric artefact: it survives replacing the quadratic with score-bin dummies and a linear spline, and future public scores carry no information about current entry or stock once the lag is controlled. Turning to the type of private sector a district gets, the quality margins trace the mirror image. Enrolment per private school, the share of private teachers with a graduate degree, and the private infrastructure index are all U-shaped in public quality, reaching their minima in precisely the low-to-middle range where entry is most intense. Private schools' own ASER scores, by contrast, rise monotonically with public scores. Taken together, the extensive and intensive margins describe a vertically differentiated market: weak public schools are colocated with many small, modestly equipped private entrants, while strong public sectors sustain fewer, larger and better-staffed private schools.

(*JEL*: I21, I25, L11, L15, M13)

Keywords: Private schools, Public schools, Evolution of market for education

1 Introduction

Private schools play an increasingly prominent role in India's education landscape. Over the last two decades, the rapid growth of low-fee private schools has reshaped local schooling markets, expanded household choice, and generated a complex set of interactions between public and private provision. This expansion has been accompanied by a persistent debate about the relative quality of public and private schools, and its implications for efficiency and equity of schooling markets. Early work argued that

observed differences in test scores largely reflect selection, with private schools enrolling students from households with more favorable inputs and backgrounds (Kingdon, 1996; Kingdon, 2007; Chudgar and Quin, 2012). A parallel strand of research, however, finds that private schools often generate learning outcomes comparable to public schools at substantially lower cost, suggesting differences in efficiency rather than student composition (Muralidharan and Sundararaman, 2015; Singh, 2015). At the same time, concerns about the quality of public schooling have remained central to policy debates, with substantial variation across districts in both the level and trajectory of public learning outcomes. While the literature has focused primarily on comparing levels of quality across sectors, much less is known about how changes in public school quality reshape private school behavior and market structure over time. Understanding how private schools respond to improvements or declines in the public option is therefore essential for interpreting the joint evolution of schooling markets and for designing policies that improve access, efficiency, and learning.

This paper asks the question: how does the quality of public schools shape the scale, composition, and quality of the private schooling sector? While households choose between public and private schools based on price, location, and perceived quality, private schools make entry and investment decisions in response to expected demand and the cost of quality provision. Public school quality enters this equilibrium not only as a determinant of relative quality, but also as a factor that shapes household willingness to pay, the minimum quality threshold required for private schools to remain viable, and the profitability of new entry. Yet despite its centrality, the empirical relationship between public sector performance and private sector behavior remains poorly understood.

We make progress by constructing a rural district-year panel that links ASER measures of learning among public school children with a reconstructed panel of the universe of schools recorded in UDISE, covering 2012-2017. The ASER measures provide standardised, comparable, widely disseminated indicators of basic reading and arithmetic proficiency across the country, which we treat as proxies for the effectiveness of local public schools. UDISE allows us to track the stock of private schools, the gross entry of new private schools, enrolment per school, and quality margins such as teacher qualifications and school infrastructure. Because raw administrative counts are contaminated by late first-filings, identifier churn and re-registration following district bifurcation, we work throughout with a reconstructed panel in which a school is counted from its establishment year to its last appearance in the directory. The districts bifurcated after the 2011 Census are dropped to prevent spurious double counting, and the final estimation sample covers 538 rural districts.

We begin by documenting a robust non-linearity on the extensive margin. Across specifications we find an inverted-U relationship between lagged public school quality and private school activity. The stock of private schools turns at a score of approximately 3.09, close to the twenty-fifth percentile of the score distribution, and gross entry turns at approximately 2.95, between the tenth and fifteenth percentiles. For the majority of district-years, therefore, better public learning predicts a smaller private sector, while

the rising segment is confined to the bottom of the distribution. Entry responds more sharply than the stock, which is consistent with entry, rather than exit, driving the stock relationship. The pattern is present at the primary level, weaker at the middle level, and absent at secondary; within each level the turning point sits at roughly the twenty-fifth percentile of that level's own score distribution.

We then examine the kind of private sector each district gets. Here, the quality margins trace the mirror image of the entry curve. Enrolment per private school, the share of private school teachers holding a graduate degree, and the private school infrastructure index are each U-shaped in lagged public quality, with minima at scores of roughly 3.13, 3.26 and 3.14 respectively, i.e., private schools are smallest and most thinly resourced in precisely the low-to-middle range in which entry is most intense, and both larger and better equipped at the higher end. The private-to-government enrolment ratio has a positive and *linear* conditional slope with no detectable curvature: relative private enrolment is higher where government schools are stronger, which together with the size margin implies fewer but larger private schools in those districts. Private schools' own ASER scores rise monotonically and stably with government scores, by approximately 0.25 points per point, with no curvature. Public and private learning are thus strong district-level complements, though the reduced form cannot separate sorting, common local inputs, and competition as the driving mechanisms.

Identification comes from variation in public school quality across districts *within* a state and year. State-by-year fixed effects absorb all contemporaneous state-level policy, including recognition drives, RTE implementation, budget changes and curriculum reform; a rich set of lagged district controls such as parental education, the number and normalised enrolment of government schools, the government infrastructure index and nightlight intensity, together with Census 2011 population levels and their interactions with a trend, isolate public school performance from broader local development. Standard errors are clustered at the district level. We do not include district fixed effects in the main specification: in a six-year panel they absorb almost all the variation in the outcome, and we report that variant in the appendix. The use of lagged, government-only test scores mitigates the concern that private entry mechanically depresses contemporaneous public school performance through cream-skimming; we bound this concern further with a leads placebo, which the quantity margins pass and the teacher-quality margin does not. Robustness exercises confirm that the inverted-U is not sensitive to functional form, panel construction, or the choice of score. It is also not an artefact of treating an ordinal ladder as a cardinal scale: the same shape appears when learning is measured by the district proportion of government-school children clearing each ASER rung — the form in which ASER itself reports — and when the score is replaced by nothing more than its within-year quintile, so that only the ordering of districts is used. Because a percentile is invariant to any monotone re-labelling of the rungs while a score level is not, we state turning points in percentile terms throughout and treat the cardinal value as a convenience.

This paper contributes to a growing body of work on public-private interactions in education and the organization of schooling markets. A large literature emphasizes the role of information frictions and parental

responses to perceived school quality in shaping enrollment patterns (Andrabi, Das, and Khwaja, 2017). Complementary theoretical work models schooling markets as vertically differentiated systems in which households sort across providers and schools choose quality and scale in response to competitive pressures (Epple and Romano, 1998; Epple, Romano, and Sieg, 2006). Related empirical research highlights the importance of local market structure and spatial competition, showing that school behavior depends not only on public-private distinctions but also on the density, proximity, and quality distribution of nearby schools (Hoxby, 2000; Neilson, 2013). We examine how private schools adjust their entry decisions, scale, and quality investments in response to local changes in the quality of the public option. By jointly studying extensive margins (private school entry and the stock of private schools) and intensive margins (enrolment per school, teacher qualifications, and infrastructure) we provide new evidence on how private providers position themselves as local competitive conditions evolve. Our results underscore the importance of viewing schooling markets as equilibrium systems in which improvements in public provision reshape private incentives not only through direct public-private competition, but also through changes in local market structure and competitive pressure among private schools themselves.

The remainder of the paper proceeds as follows. Section 2 provides background on India's schooling system and descriptive trends in public and private sectors. Section 3 describes the data sources and variable construction. Section 4 outlines the empirical strategy. Section 5 presents the main results on private school entry and scale. Section 6 examines intensive-margin responses and discusses mechanisms. Section 7 reports dynamics and placebo tests, Section 8 heterogeneity, and Section 9 robustness. Section 10 concludes with implications for policy and schooling market design.

2 Background

The Indian Education system is vast and reflects persistent tension between the government's mandate of universal education, with the *Sarva Shiksha Abhiyaan* programme, and rapid concurrent sprouting of private schools. The relationship between public school quality and private schools proliferating can be understood as a co-evolutionary process. This paper contributes to this understanding by empirically testing how changes in public school quality affect private school entry across districts.

Early work by Kingdon (2007) and Chudgar (2009) established that the growth of private schools in India is not confined to urban or affluent populations. Even in rural areas, low-fee private schools have emerged as viable alternatives to the public option, driven by parental dissatisfaction with public school deficiencies such as teacher absenteeism, multi-grade classrooms, and limited accountability.

Pal (2010) shows that private schools are more likely to locate in villages with better local public infrastructure, which includes, among other things, access to piped water, electricity, phones and a post office. This is shown to be driven in part by greater demand from households with greater education and

from those belonging to higher castes, and these are unequally distributed across villages. Pal (2010) also documents higher average pass rates in villages with better public infrastructure and private school presence, but notes an insignificant impact of greater private school presence on the pass rates of existing public schools in these villages. More recent work like Bagde et al. (2022) also suggests that private expansion has led to little to no improvement in public outcomes.

The "push" and "pull" of public education demonstrate not just the inner workings of the market, but social negotiations between the state and private players. Debates on school quality in India often hinge on whether private provision outperforms public schools once we adjust for student background and sorting. Kingdon (1996) finds, in urban India, that raw score comparisons are misleading as they do not account for endogenous selection, and estimates achievement functions that correct for selection and home inputs before attributing differences in quality or efficiency to management type. This exercise shows why private schools may sometimes look better in simple comparisons yet converge toward parity when selection is controlled for. More recently, Central Square Foundation (2020), in a state of the sector report, underscores the sheer scale of private enrollment and emphasizes that learning advantages are not consistently higher after accounting for student background; it argues for outcome-focused regulation rather than input compliance.

With the rapid expansion of the private sector in school education, including the proliferation of low-fee private schools, parents have been choosing in relatively low-information markets. Field evidence from four states (Azim Premji University Research Group, 2018) shows that families greatly value characteristics of schools such as English-medium, perceived discipline, and safety; yet perceptions often diverge from reality (e.g., "English-medium" may be in name only, or less-qualified teachers in schools chosen for "teacher quality"). Afridi et al. (2020) conduct an informational intervention which shows that there is a re-sorting of children across schools when parents and schools randomly receive information about the test scores of all students in a local market. This intervention results in strong statistically significant improvements in average test scores, showing that school choice decisions are made with incomplete information.

Prior work on market dynamics shows that rural entry of private high schools induces sharp stratification by achievement, caste, gender, and affluence; nevertheless, large-scale administrative data show no short-to-medium-run declines for public-school students left behind, suggesting competitive pressures and resource reallocation can be offset by peer-group losses (Bagde et al., 2022). Upstream supply shocks matter too: the rollout of elite public colleges appears to catalyze local private-school entry and shift students from public to private institutions, likely via improved infrastructure and reduced travel/setup costs (Jagnani and Khanna, 2020).

Demand mechanics are central. In nearby Pakistan's dense low-fee private markets, families place a high premium on proximity (strong distance disutility), face relatively low own-price elasticities, and derive

sizable consumer surplus from having a private option at all (Carneiro et al., 2022). Inside households, interpreting expenditure "parity" requires care: gender bias can operate through the extensive margin (non-enrollment for girls) even where per-enrolled-child spending looks equal, so detection methods matter (Kingdon, 2005).

3 Data

Our empirical analysis draws on multiple administrative and geospatial datasets spanning 2011–2017. The primary source is the District Information System for Education (DISE), the annual census of all registered schools in India. DISE provides detailed information on schools—including management type, category, year of establishment, and infrastructure (such as piped water, sanitation facilities, electricity, libraries, computers, boundary walls, and playgrounds). It also includes data on student enrollment and teacher characteristics (number, qualifications, etc.). Using the School ID variable and the year of establishment, we construct a school-level panel dataset.

A complication arises because School IDs are not always stable over time: the same school may appear under multiple codes across years, or conversely, the same code may correspond to different schools. Furthermore, when a geographical unit (such as a state, district, block, or village) is bifurcated, both the parent and child units can appear, potentially leading to double-counting. To address this, we systematically eliminate both the parent and child districts in years of bifurcation, ensuring that totals are calculated without duplication at the district–year level.

Starting in 2012, DISE integrated primary and middle school data with secondary school data and upgraded its data collection infrastructure, creating the Unified District Information System for Education (UDISE). Since then, self-reporting by private schools has improved steadily, and by 2017, UDISE achieved full coverage of the school universe (Ministry of Education, 2022). Leveraging this coverage, we construct procedures that yields two versions of a temporally consistent school-year panel for private schools. In both constructions, schools are identified by their reported school codes and expanded into school-year observations based on establishment information.

In the first construction (the union panel), a private school is treated as active in year t if its reported estabblishment year is at most t. Once a school appears in a year, it is carried forward through 2017. This construction gives us an upper bound on the stock oof private schools by abstracting from exit mechanisms and is useful for assessing the role of entry and identifier instability. In the second construction (the panel with exits), we additionally restrict a school's active years to those up to its last observed year in UDISE. Permanent disappearance from the administrative records is therefore interpreted as market exit. This yields a comprehensive, non-repeating panel in which each school is observed from its year of establishment (or first appearance) until its final recorded year.

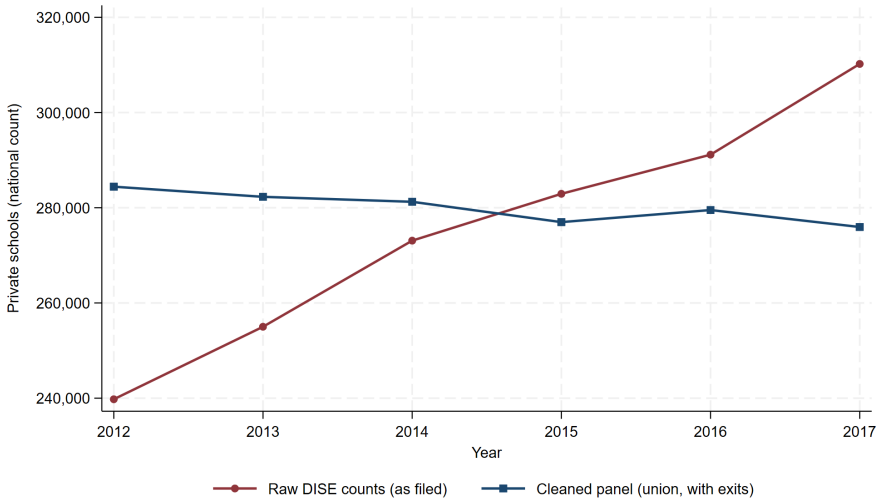


Figure 1: National private school count, 2012–2017: raw DISE filings against the cleaned union-with-exits panel. Raw as-filed counts rise from 239,776 to 310,211 (+29%); the cleaned operating stock falls from 284,425 to 275,956 (−3%). The series cross in 2015 — the reconstruction back-fills late first-filers, so it exceeds the raw count early, and nets out administrative re-registration, so it falls below the raw count late.

We aggregate to the district–year level to compute our main dependent variable: the total number of private schools. Unless otherwise stated, all main results use the panel with exits. Figure 1 contrasts raw and cleaned national private-school counts over time. Raw administrative counts increase sharply, reflecting mechanical inflation from identifier changes and bifurcations. In contrast, the cleaned series exhibits substantially lower volatility and evolves gradually, indicating that the cleaning procedure restores temporal coherence without imposing smoothness by construction. In the cleaned data, fluctuations are evident—most notably a sharp dip in 2015 followed by a rebound in 2016—the overall trajectory is downward. Between 2012 and 2017, the total fell from roughly 2,84,425 to 2,75,956 schools, a net loss of about 8,469 private schools (a decline of approximately 3%). The cleaned series should not be interpreted as the true level of private schools, but as a temporally consistent measure that removes mechanical inflation over time.

Figure 2 shows the distribution of district-level year-to-year changes in private-school counts. Raw administrative data exhibit substantially greater dispersion and a positive shift in the distribution, consistent with mechanical inflation from identifier changes and bifurcations. In contrast, the cleaned data display a much tighter distribution centered near zero, indicating improved temporal coherence without eliminating genuine entry and exit. Outside values are excluded from the figure for visual clarity.

We matched the annual census of Indian schools, the UDISE, 2012-2017 to the Population Census (2011) for village-level amenities data. We also match the Visible Infrared Imaging Radiometer Suite (VIIRS)

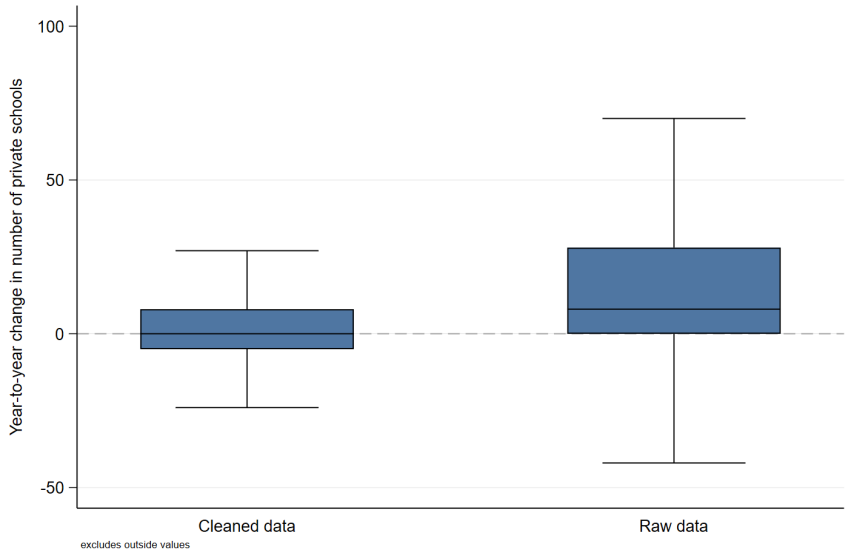


Figure 2: Distribution of district-level year-to-year changes in the private school count, raw against cleaned. Outside values omitted for legibility.

nighttime lights data, which was processed to create annual district-level measurements.[1] Spatial data from the Census and the World Bank were also matched. The ASER dataset (2011–2017) was merged with the remaining datasets for district-level educational indicators, such as reading and mathematics learning outcomes. As ASER is conducted only in rural areas, the analysis sample is correspondingly restricted to rural districts. ASER was not carried out in 2015 and 2017. So, we include 2011 data as well to be able to use lagged scores for data in 2012.

The resulting panel dataset covers 538 districts over the 2011–2017 period. Table **??** summarizes sample attrition across data construction stages. The excluded column reports the number of districts dropped at each stage relative to the immediately preceding sample, rather than cumulative exclusions. The final dataset contains harmonized information on educational infrastructure, learning outcomes, economic activity, and spatial characteristics, and is well-suited for analyzing district-level educational dynamics.

[1] Monthly sub-district-level data was aggregated and harmonized to align with Census 2011 district boundaries.

4 Private Schools

This section provides descriptive evidence on the size, composition, and variation of schooling markets across districts in India. We document key sources of variation, and motivate the empirical specifications that follow. Throughout, we emphasize heterogeneity across districts rather than national averages, reflecting the localized nature of schooling markets.

4.1 Size and Composition of Schooling Markets

We consider the relative size of public and private schooling sectors across districts first. Public schools account for the majority of total enrollment in nearly all districts, but private schools constitute a substantial and highly heterogeneous share of local schooling markets. In the median district-year, private schools represent approximately percent of all schools, with the interquartile range spanning from to percent. In some districts, private schools are nearly absent, while in others they form a sizeable fraction of total provision.

Figure plots the distribution of the number of public and private schools across districts. While public schools dominate in absolute numbers, the dispersion in private school counts is considerably larger. This dispersion persists even after normalizing by district population or total enrollment, suggesting that variation in private school presence reflects more than differences in district size alone. Instead, it points to differences in local demand conditions, institutional environments, and the perceived quality of public provision.

Private school presence also varies systematically across states, but substantial heterogeneity remains within states. Districts in the same state—and subject to the same education policies and regulatory frameworks—exhibit markedly different private school shares. This within-state variation motivates our empirical strategy, which exploits differences in public school quality across districts within the same state and year.

4.2 Variation in Public School Quality

Public school quality, measured using district-level averages of ASER reading and mathematics scores, exhibits substantial variation across both space and time. Figure plots the distribution of lagged public school ASER scores across districts. While average learning levels remain low in absolute terms, the distribution is wide, with meaningful differences between districts at the lower and upper tails.

Importantly, public school quality is not static. Over the period 2012–2017, many districts experience nontrivial changes in average ASER scores, though the magnitude and direction of these changes vary considerably. Figure illustrates this heterogeneity by plotting district-level changes in public school ASER scores over time. Some districts exhibit steady improvements, others stagnate, and a smaller subset experiences declines.

These changes in public school quality are not perfectly correlated with observable indicators of economic development. For example, while districts with higher nightlight intensity tend to have higher average ASER scores, the correlation is far from one-to-one. Conditional on state fixed effects, substantial variation in public school learning outcomes remains unexplained by standard proxies for income or demographic composition. This residual variation forms the basis of our empirical analysis.

Taken together, these patterns indicate that districts differ meaningfully in both the level and trajectory of public school quality. This variation provides a natural setting for examining how private schools respond to changes in the quality of the public option within localized schooling markets.

5 Empirical Model

$$Y_{dt} = \alpha_0 + \alpha_1 \cdot T^g_{dt-1} + \alpha_2 \cdot T^{g^2}_{dt-1} + \boldsymbol{\alpha_3} \cdot \boldsymbol{W_{dt-1}} + \alpha_4 \cdot X_{dt} + \boldsymbol{\alpha_5} \cdot \boldsymbol{Z_d} + \boldsymbol{\alpha_6} \cdot \boldsymbol{Z_d} \cdot t + \gamma_{1_s} + \gamma_{2_t} + \gamma_{3_{st}} + \epsilon_{dt} \quad (1)$$

The marginal effect of public school quality is $\alpha_1 + 2\alpha_2 T^g_{dt-1}$ and the turning point is at $-\alpha_1/2\alpha_2$.
The main outcome variables of interest that we examine are: (1) Y_{dt}, the number of private schools in a district d in year t, and (2) E_{dt}, the number of new private schools in district d at time t, i.e., entry in district d in year t. T^g_{dt-1} is the average combined reading and math ASER score for public school students of district d in year t-1. In specification (1), $\alpha_1 + 2 \cdot \alpha_2 \cdot T^g_{dt-1}$ is the marginal effect of an increase in average public school students' ASER scores on the number of private schools. Similarly, in specification (2), $\beta_1 + 2 \cdot \beta_2 \cdot T^g_{dt-1}$ is the marginal effect of an increase in average public school students' ASER scores on the number of new private schools. The marginal effects are themselves functions of the average public school students' ASER scores due to the quadratic specification.

$\boldsymbol{W_{dt-1}}$ is a vector of lagged controls at the district-year level, which includes variables such as average parental education, total number of public schools, average total enrollment in public schools, average infrastructure index[2] in public schools, and average night light intensity. These variables can presumably affect both ASER score averages of public school students (either through the education production function or through income effects) and the number of private schools in a district by their effect on demand for private schooling. X_{dt} is the number of blocks and villages in district d in year t; it is added as a control for the size of the district, and also to account for the measurement error introduced by double counting of schools due to bifurcation of blocks and villages[3]. $\boldsymbol{Z_d}$ is a vector of time-invariant district-level controls such as the value of an infrastructure index[4] for district d in the census year 2011,

[2]Calculated using principal components analysis of amenities available in each school as reported in DISE. These amenities include the presence of drinking water, boundary walls, computers, functioning electricity, ramps, classrooms in good condition, toilets for boys and girls, computer-aided learning, medical checkups and meals in schools.

[3]When there is an administrative bifurcation of blocks or villages, there is a change in the unique identifier (school code) of schools in DISE. As a result, two different school codes may represent the same school in different years due to a change in the administrative division it is part of.

[4]Calculated using Census variables on connectivity, health and sanitation.

the percentage of population of district d that is rural in 2011, the percentage of population of district d that is SC in 2011 and the percentage of population of district d that is ST in 2011. Trends of these variables are also included in the specification.

γ_{1_s} and δ_{1_s} represent state fixed effects, γ_{2_t} and δ_{2_t} represent year fixed effects, and $\gamma_{3_{st}}$ and $\delta_{3_{st}}$ represent state-year fixed effects in the respective equations. These fixed effects partial out the effect of confounders that vary at both the state and year level, such as state education policy. In the appendix, we display the results of the main specification with more general district fixed effects which control for all possible district-level confounders. We find that district fixed effects explain a high proportion of the variation in the dependent variables, with an R^2 value of 0.99 when district fixed effects are included. Since the panel spans only six years of data, temporal variation within districts is limited. Including district fixed effects thus leaves very little variation for identifying the marginal effect of average public school students' ASER scores on private school numbers and entry. Thus our main specification uses district-year level excess variation in average public ASER scores above state-year averages, in addition to controlling for several district-level variables (both time-invariant $\boldsymbol{Z_d}$ and time-variant $\boldsymbol{W_{dt-1}}$) to achieve identification.

The use of lagged values of average ASER scores instead of contemporaneous values is done in order to avoid feedback effects. It is well-documented that test scores of students in private schools are higher than of those in public schools, primarily through selection (add references). As a result, having a greater number of private schools and greater enrollment in private schools can presumably affect district-wide average public school scores. The effect may be negative or positive depending on whether the selection of more able students in private schools leads to worse peer effects for public school students, or if competition from private schools and greater number of resources per child in public schools boosts test scores for public school students. Using past values of district-wise ASER average score ensures that the correlation observed is not driven by reverse causality. Moreover, in a recently published paper, Bagde et al. (2022) show that the entry of private secondary schools into 240 Indian villages (which were previously served by public schools only) has no effect on the scores of public school students. Similarly, Michaud-Leclerc (2022) also finds, in the context of rural Pakistan, no negative impact of private school expansion on test scores of public school students in grades 3-5, and no negative impact in the long term either. These results dispel any remaining concerns about the possibility of reverse causality affecting our estimates.

6 Results

6.1 Number of Private Schools (Stock and Entry)

Table 2 presents our main results, with the 'total number of private schools in district d at time t' and the 'number of new private schools in district d at time t (entry)' as outcome variables. Column 2 shows an insignificant effect of public school quality on the number of private schools - the linear specification, however, masks heterogeneity. The third column, with a quadratic specification, shows that there exists

a statistically significant inverted U-shaped relationship between public school quality and the number of private schools. At low levels of public school quality (measured by the lagged average ASER score of students in public schools), there is a positive correlation between public school quality and the number of private schools. At higher levels of public school quality, the relationship reverses. In other words, in districts where the public option for schooling produces low average test scores, more private schools compete as the quality of public schools increases. In districts with high average test scores for public schools, incremental increases in test scores are associated with fewer private schools in the market. The marginal effects and predicted values of outcome variable for different values of lagged average ASER scores corresponding to the specification in column 3 are plotted in Figure 6. These patterns reveal that the private schooling market is competitive in districts where public schooling is wanting, and are fewer in number when the existing public option is strong.

Column 6 of Table 2 suggests that there is a similar pattern in the relationship between public school quality and the number of *new* private schools in a year, i.e., entry. Figure 7 plots the marginal effects of the lagged average ASER scores of public schools on the entry of new private schools and the predicted entry of new private schools at different levels of the lagged average ASER scores of public schools. The similar patterns in Figures 6 and 7 suggest that the effect on total number of private schools may be driven by entry. [Enter reference about Kala Krishna's latest paper about regulatory environment making exit of firms difficult – implying exits may not respond to public school quality as flexibly as entry of new private schools might.]

Taken together, these results suggest that changes in the number of private schools, and especially in new private school entry, are responsive to the quality of public education. We next examine whether the relative scale of private schooling—measured by the ratio of enrollment in private schools to that in public schools—shows a similar pattern. This provides a sense of how household demand shifts across school types in response to public school performance.

6.2 Ratio of enrolment in private schools to enrolment in public schools

Table 35, Table 11 and Figures 16 and 15 present these results. The outcome variable is the ratio of normalized enrollment in private schools (and in new private schools for the entry specification) to normalized enrollment in public schools in district d at time t. The second panel of Figure 15 plots the predicted values of this ratio against lagged average ASER scores of public school students, while the first panel (Figure 15) shows the corresponding marginal effects.

The point estimates trace an inverted-U pattern similar to that observed for the number of private schools and for entry, though the confidence intervals are wider and encompass zero over much of the range. This statistical imprecision likely reflects the short time span of our data and the fact that aggregate enrollment adjusts more slowly than the extensive margin of entry. If school fees and switching costs for households are sticky within a few years, when private profitability conditions change, student reallocation across

sectors may be slow to adjust. Despite the broad confidence bands, the central estimates are economically consistent with the earlier findings: at low levels of public school quality, improvements in public test scores are associated with a rising ratio of private to public enrollment, while at higher levels of public quality, the slope turns negative. This suggests that private enrollment expands where public schooling is weak but contracts once the public option becomes sufficiently effective. The results, while imprecisely estimated, reinforce the interpretation that the private sector's footprint in schooling markets reflects the relative profitability of operating vis-à-vis public schools.

7 Cost Margins and Private Investments

7.1 Quality of Private Schools

Table 12 presents results from our main specification where the dependent variable is the average ASER score of students enrolled in private schools in district d at time t. The key explanatory variable is the lagged average ASER score of students enrolled in public schools, along with its quadratic term, and we continue to include district-level and district-year-level controls, as well as year, state, and state-year fixed effects. In contrast to earlier results on private school quantity and entry, we find that private school quality exhibits a monotonically increasing—and statistically significant—relationship with public school quality. The estimate is strikingly stable across specifications, at roughly +0.25 points of private score per point of public score, and the quadratic term is small and statistically indistinguishable from zero: unlike every other margin we examine, this relationship shows no curvature. Both the marginal effects and the predicted values plotted in Figure 17 are therefore close to linear.

This pattern suggests that public and private learning are strong complements at the district level. Improvements in public schooling may raise the overall equilibrium quality of education in the local market, by shifting parental expectations, intensifying reputational competition, or increasing the opportunity cost of low-quality provision among private schools. Equally, both sectors may simply draw on the same local inputs—teacher supply, household literacy, and the general level of development—or the same households may sort into both. The reduced form cannot separate these channels. What it does establish is that the districts where the public sector teaches children to read are also the districts where the private sector does, even though those are the districts with the fewest private schools.

7.2 Teacher Quality

Table 13 presents the results when the outcome variable is the proportion of teachers in private schools with at least a graduate degree in district d at time t. The key independent variable is the lagged average ASER score of students enrolled in public schools, entered with both linear and quadratic terms, alongside the standard set of district-level and district-year-level controls, and year, state, and state-year fixed effects. The estimates reveal a statistically significant *U-shaped* relationship. The linear term is

−0.291 (SE 0.093) and the quadratic +0.045 (SE 0.014), both significant at the one per cent level, placing the minimum at a score of approximately 3.26. Bivariately the relationship looks positive (+0.022, not significant once controls enter), so the shape emerges only conditional on the control set and state-by-year fixed effects. Both the marginal effects and the predicted values plotted in Figure 18 show that the graduate-teacher share falls with public quality over the lower part of the range, bottoms out just above the median district, and rises thereafter.

This is the mirror image of the entry curve rather than a monotone upgrading response, and it is the shape the mechanism has to explain. Where public schools are weakest, private entry is at its most intense and the schools that enter are staffed with the least-qualified teachers—consistent with a thin, low-cost, high-turnover fringe serving unmet demand. Where public schools are strongest, entry is sparse but the private schools that operate employ better-qualified teachers, consistent with survival requiring differentiation on a margin parents can observe. The minimum sits close to where entry has already turned down, so the two curves are not independent descriptions of the market but two readings of the same repositioning.

One caveat belongs here rather than in the robustness section. This margin does not pass the lead placebo reported in Table 19: future public scores predict private teacher credentials nearly as well as past scores ($p = 0.013$ on the lead terms). The U-shape should therefore be read as a persistent district-level association between public and private teacher quality, not as a private-sector response to changes in public quality.

7.3 Infrastructure Index

Table 15 presents the results where the dependent variable is the private school infrastructure index in district d at time t, constructed as a standardized composite (using Principal Components Analysis) of key infrastructure indicators such as availability of classrooms, toilets, electricity, boundary walls, computers etc. The main explanatory variable remains the lagged average ASER score of students enrolled in public schools, included in both linear and quadratic form, and the regression continues to control for the full set of district-level and district-year-level covariates, as well as year, state, and state-year fixed effects. The estimates trace the same U-shape as teacher quality. The linear term is −0.762 (SE 0.349) and the quadratic +0.121 (SE 0.052), both significant at five per cent, placing the minimum at a score of approximately 3.15. As with teacher quality, the bivariate relationship is strongly positive (+0.641, significant at one per cent) and collapses once the controls enter, so the raw positive correlation reflects the general level of local development rather than anything specific to public school performance. Figure 20 plots the marginal effects and predicted values.

Taken with the teacher-quality and school-size results, this completes the mirror image: all three of the margins on which a private school can be said to be better resourced—how many children it teaches, how well qualified its teachers are, and what physical facilities it has—reach their minimum in the same low-to-middle band of public quality, between roughly 3.1 and 3.3, which is precisely the band in which entry

is most intense. Districts with the weakest public schools do not simply have more private schools; they have more private schools of a systematically thinner kind. Districts with the strongest public schools have fewer private schools, but those are larger, better staffed and better equipped.

The natural reading is vertical differentiation. When the public option is weak, the quality a private school must offer to attract a paying family is low, so entry is cheap and the entrants are correspondingly modest. As the public option improves, that threshold rises: entry becomes unattractive, and the schools that remain viable are those able to bear the cost of visible quality. What the reduced form cannot settle is whether individual schools upgrade or whether the composition of survivors changes, since we observe districts rather than schools over time.

8 Dynamics and Placebo Tests

Two questions arise naturally from a specification in lagged scores. How long is the lag between a public learning signal and observed entry? And do *future* public scores predict current entry, as they would if private entry were mechanically changing the composition of government school test-takers?

Table 17 adds the second lag of the score. The first lag carries the result: the inverted-U at $t-1$ is strong in every column, while the second lag on its own is flat ($p = 0.20$). Entered jointly, however, the second lag is significant ($p = 0.003$) with signs opposite to the first, and the lag-one coefficient rises from 62.9 to 82.7. We read this as partial mean reversion: entry responds to the level of public quality and to recent movement in it, consistent with the levels-versus-changes result in Table 3. The cumulative effect remains an inverted-U, peaking at approximately 2.78.

Table 18 reports the leads placebo. Adding the lead of the score leaves the lagged inverted-U intact and the lead terms are jointly zero ($p = 0.88$ for the linear term, $p = 0.34$ with curvature). Column (4) shows why the test must be run conditional on the lag: dropping the lag, the lead reproduces the lagged coefficients almost exactly, because the correlation between T^g_{t-1} and T^g_{t+1} is 0.86. Figure 22 plots the linear-in-score coefficient at each horizon, each on its own maximal sample: the signal is present before entry and absent after it.

Table 19 extends the placebo across margins, and the result is not uniform. The quantity margins clear it: future government scores carry no information about current entry ($p = 0.34$) or stock ($p = 0.36$). The graduate-teacher share does not clear it ($p = 0.013$), with future scores predicting private teacher credentials nearly as well as past scores. That U-shape should therefore be reported as a persistent district-level association rather than a response to public quality. The private-score column is uninformative: it collapses to a single cross-section of 468 districts and the lagged coefficient itself falls to insignificance, so that comparison requires a different design. With district-year data we bound this concern on the extensive margin and fail to bound it on teacher quality.

9 Heterogeneity

A vertical differentiation account makes a sharp prediction: where households can pay more, the marginal entrant's break-even quality should be higher, so the entry peak should shift *right*. We test this by interacting both score terms with an above/below-median dummy for each of seven baseline district characteristics — nightlight intensity, parental education, SC share, ST share, rural share, graduate-or-above teachers per government school, and the share of government teachers with a graduate degree — measured at the first non-missing value up to 2013, or from the 2011 Census where time-invariant. Tables 11 and 11 report the interactions; Tables 11 and 11 re-estimate the whole specification separately within each group, so that every coefficient, not only the score terms, is free to differ.

The prediction fails along the ability-to-pay dimensions. For nightlights the entry turning point moves from 2.949 below the median to 2.918 above it, and the interactions are jointly insignificant ($p = 0.32$). For parental education it moves from 2.936 to 2.803, again jointly insignificant ($p = 0.91$). Both point estimates run in the wrong direction for vertical differentiation, and neither is distinguishable from zero. Richer districts are larger markets at every level of public quality, but the level of public quality that maximises entry is the same in rich and poor districts. Market size shifts the height of the curve; it does not relocate its peak.

Two dimensions do bite, and neither is about income. Where the ST share is above the median, the entry peak sits materially further right: 3.183 against 2.642 in low-ST districts, with the interactions jointly significant at $p = 0.009$. The stock shows the same displacement, 3.323 against 2.718 ($p = 0.033$). Where a larger share of government teachers hold a graduate degree, the entry peak shifts the other way, from 3.127 to 2.485, with both interaction terms individually significant at five per cent and jointly at $p = 0.071$. For the stock, the rural-share split is the strongest of the seven ($p = 0.006$).

How to read this. The two dimensions that move the peak are supply-side or structural rather than measures of what households can pay. High-ST districts are the ones where private entry continues to respond to public quality furthest up the score range, which is consistent with private provision arriving later and from a lower base in those markets. The government teacher-qualification result points the same way from the opposite direction: where the public sector has already absorbed the locally available qualified teachers, the private sector's entry window closes earlier. Both suggest that what shifts the threshold is the availability of the inputs a private school needs, not the depth of local demand — which is awkward for a pure vertical-differentiation story and easier to reconcile with entry into unmet demand at a roughly common quality threshold, modulated by input supply.

Two limitations should be stated plainly. First, this is a test of low power: with roughly 530 district clusters and a score standard deviation of 0.45, shifts of a tenth or two of a score point are below the resolution of district-level data, so the null results along nightlights and parental education bound the effect rather than reject it. Second, the intended inputs were not all available. The NSSO 68th-round teacher file carries no state identifier and several district names repeat across states, so a name-only merge was not reliable; DISE teacher measures stand in for it. Census education of the 30-and-over population is not in the analysis dataset, so ASER parental education serves as the education dimension. We also do

not report the above-median parental-education turning point for the stock, which the quadratic places at 23.0 — far outside the observed score range of 1.83 to 4.53, and an artefact of a nearly flat quadratic rather than a meaningful peak.

10 Robustness

10.1 Is the result an artefact of treating an ordinal scale as cardinal?

The ASER measure is a ladder, not a scale. A child is recorded at the highest of five rungs she can clear — nothing, letters, words, a Standard I text, a Standard II text on the reading side, and the arithmetic equivalents — and our regressor averages those rung numbers across government-school children in a district. That average is only meaningful if the rungs are equally spaced, which nobody claims. Because our central result is a *turning point* in that average, the objection is pointed: the location of a peak in a rescaled variable need not survive rescaling, and in principle a monotone re-labelling of the rungs could manufacture or destroy curvature. We therefore re-estimate the headline entry and stock specifications using two definitions of learning that use no cardinal information at all.

The first replaces the average with the district *proportion* of government-school students clearing a given rung: subtraction or above, division, reading a Standard I text or above, and reading a Standard II text. These are exactly the quantities ASER itself reports, and each is a proportion rather than a scale score, so no spacing assumption is required. The second is a pure rank check: we replace the score with dummies for its within-year quintile, discarding everything except the ordering of districts within each year. Controls, fixed effects, clustering and sample are held fixed throughout, at $N = 1{,}985$.

Tables 29 and 30 report the proportions. The result is unchanged. Against a baseline entry inverted-U of +49.8 and −8.5 (both significant at one per cent, turning point 2.92), every proportion enters negatively and significantly at one per cent in the linear specification, with coefficients from −19.7 to −24.5, and every quadratic reproduces the inverted-U. The shape is sharpest for the reading Standard I rung, at +61.4 (five per cent) and −76.5 (one per cent), with a turning point at a proportion of 0.40 against a district mean of 0.57 — so, as with the cardinal score, the majority of districts sit on the declining segment. The stock reproduces the same pattern at conventional five to ten per cent significance.

Tables 31 and 32 report the rank-only version. Relative to the bottom quintile, entry is flat in the second quintile and then falls monotonically, reaching −5 to −8 in the fourth quintile and −6.5 to −10.4 in the fifth, significant at one per cent across all five measures of learning. The stock declines the same way, with fifth-quintile coefficients of −48 to −137. There is no sign flip on any measure, for either outcome, under either method.

The substantive conclusion is that the cardinal averaging of the one-to-five scale is not doing the work. It also suggests how the result is best stated. Claims of the form "the turning point is at a score of

3.09" are exposed to the ordinality objection in a way that claims of the form "the turning point sits near the twenty-fifth percentile of the score distribution" are not, since a percentile is invariant to any monotone re-labelling of the rungs. We therefore lead with the percentile statement throughout and treat the cardinal turning point as a convenience.

10.2 Functional form

Functional form. Tables 22–24 replace the quadratic with quintile and decile dummies in the lagged score, and with a linear spline knotted at the quintile cutpoints, holding the controls, fixed effects and clustering fixed. The shape survives binning: entry declines monotonically above the second bin, with the top two quintiles at -8.0 and -9.9 relative to the first — around half of mean entry — and the bins jointly significant ($p < 0.001$); the stock mirrors this. The rising segment is not one influential cell: the second decile exceeds the first by 5.2 entrants per year, the spline slope below 3.04 is $+10.2$ per score point, and an equal-slopes test rejects linearity ($p = 0.003$). The bins peak at the quadratic's implied turning point, 2.95 with a 90 per cent interval of $[2.67, 3.24]$. The quality margins retain the U shape under binning, dipping at the second and third quintiles and recovering at the top, though the shallow U's lose power at bin level.

11 Discussion and Conclusion

The empirical patterns documented above point to a systematic, non-linear response of private schools to changes in public school quality. While the reduced-form estimates do not identify a single causal channel, the joint behavior of extensive- and intensive-margin outcomes helps discipline the set of plausible mechanisms.

A first class of explanations emphasizes profitability and competitive pressure in vertically differentiated schooling markets. When public schools perform poorly, even modest improvements in public learning outcomes may increase overall demand for schooling without substantially improving the outside option faced by private providers. In this regime, private schools can attract students at relatively low cost, encouraging entry and expansion. As public school quality continues to improve, however, the public option becomes a closer substitute for low- and mid-quality private schools. Private enrolment per school declines, entry slows, and the marginal private school faces increasing pressure to differentiate on quality. The observed convex increases in teacher qualifications and infrastructure investments are consistent with this mechanism, reflecting costly quality upgrades undertaken by private schools that remain in the market.

A second possibility is a market-composition mechanism, whereby improvements in public school quality selectively displace low-quality or informal private providers. In this view, the decline in private entry and scale at higher public quality levels reflects exit or non-entry of marginal private schools, while

the remaining private schools are higher-quality by construction. Two features of the results, however, weigh against pure composition effects. First, quality-related outcomes rise smoothly rather than discontinuously with public school quality, suggesting incremental adjustment rather than discrete turnover. Second, similar patterns emerge within primary and middle school subsamples, which differ in baseline private-school composition.

A third explanation centers on household sorting and demand reallocation. As public schools improve, households—particularly those at the margin between low-fee private schools and public schools—may re-enroll in the public sector. Such sorting would reduce private enrolment and entry while increasing the average quality demanded by remaining private-school households, inducing higher investment in teachers and infrastructure. While sorting likely plays a role, this mechanism alone cannot account for the sharp decline in enrolment per private school observed at higher levels of public quality, which points to shrinking market size rather than merely changing clientele.

Taken together, the combination of declining private enrolment per school, reduced entry at higher public quality levels, and convex upgrading along costly quality margins is most consistent with a profitability-based mechanism in which improvements in the public option compress private margins and raise the minimum quality threshold required for private schools to remain viable. While alternative mechanisms may operate simultaneously, the internal coherence of responses across margins suggests that changes in public school quality fundamentally reshape private incentives rather than simply reallocating students across sectors.

These findings have several implications for how public investments in education interact with private provision. First, they suggest that improvements in public school quality need not generate a monotonic crowding-out of private schools. Instead, public quality improvements can initially coexist with, and even stimulate, private entry and expansion when public schools are sufficiently weak. In this range, private schools appear to operate as close but imperfect substitutes, serving unmet demand and responding to rising concern about learning outcomes. Only once public schools reach a higher level of effectiveness does the competitive pressure exerted by the public option begin to compress private margins and discourage entry.

Second, the results underscore the importance of distinguishing between extensive and intensive margins when evaluating the impact of public-sector reforms. While the number of private schools and private enrolment decline at higher levels of public quality, private schools that remain in the market respond by upgrading teacher qualifications and physical infrastructure. These adjustments suggest that competition from improved public schools shifts the private sector toward higher-cost, higher-quality provision rather than simply inducing exit. From a welfare perspective, this implies that the consequences of public school improvements for private schooling cannot be assessed solely through changes in the number of private schools or enrolment shares, but must also account for changes in the quality of private provision.

Third, the non-linear nature of the response highlights the relevance of local market structure in shaping equilibrium outcomes. Districts with low public school quality appear to support a relatively large number of low-cost private providers, whereas districts with stronger public schools sustain fewer private schools that operate at higher quality and cost. This pattern is consistent with schooling markets characterized by vertical differentiation and heterogeneous providers, and suggests that public quality improvements can reallocate private supply across segments rather than eliminating private provision altogether.

Several limitations of the analysis warrant discussion. The relatively short time span of the panel limits our ability to observe long-run adjustments, particularly along slow-moving margins such as enrolment and school exit. Data constraints also prevent us from observing private school fees, closures, or within-school changes in inputs over time, which would allow for sharper discrimination between within-school upgrading and compositional change. In addition, while lagged public school test scores and rich fixed effects mitigate concerns about reverse causality and confounding trends, the estimates should be interpreted as descriptive equilibrium relationships rather than causal effects of specific policy interventions. These limitations point to promising directions for future research. Linking administrative school data with information on fees, household beliefs, and regulatory enforcement would allow for more direct tests of the mechanisms emphasized here. School-level panel data would further enable modeling of entry and exit decisions and the dynamics of quality upgrading. Finally, exploiting quasi-experimental variation in public school quality — such as policy-driven changes in teacher deployment, or school inputs, or public school rationalisation — would provide sharper causal estimates of how private providers respond to public-sector improvements.

In conclusion, this paper provides new evidence on how changes in public school quality shape the structure and behavior of private schooling markets in India. Rather than inducing uniform crowding-out, improvements in public schooling generate a non-linear set of responses that operate through both entry decisions and costly quality adjustments. These findings highlight the importance of viewing public and private schools as part of a single, interdependent system, and suggest that policies aimed at improving public school quality can have nuanced and heterogeneous effects on private provision. Designing effective education policy therefore requires accounting not only for direct effects on public schools, but also for the equilibrium responses of private actors operating alongside them.

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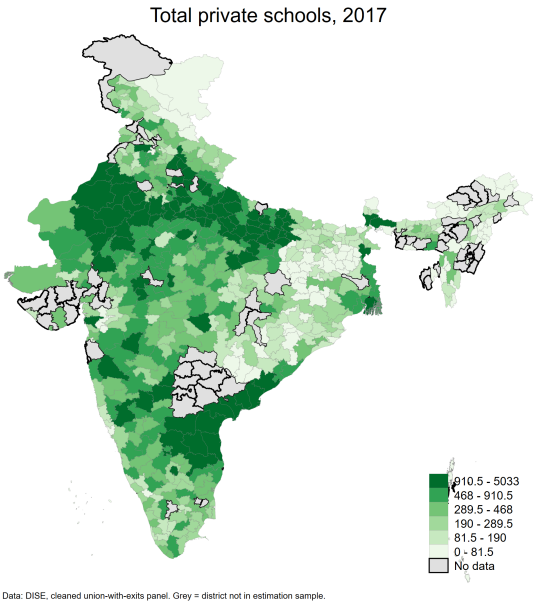
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Figures and Tables

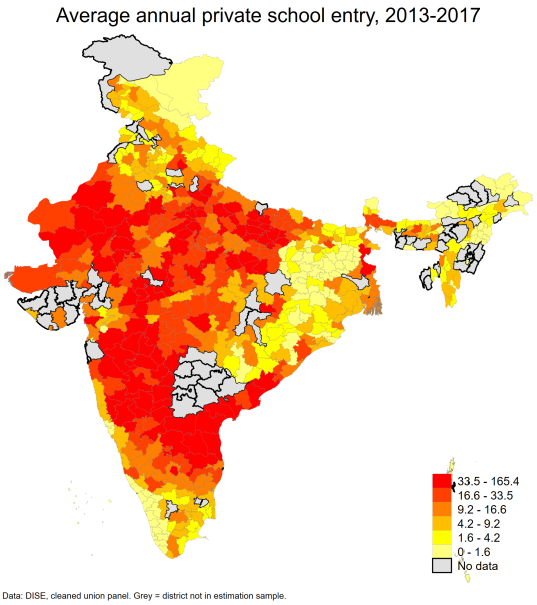
Descriptives

Table 1: Summary Statistics

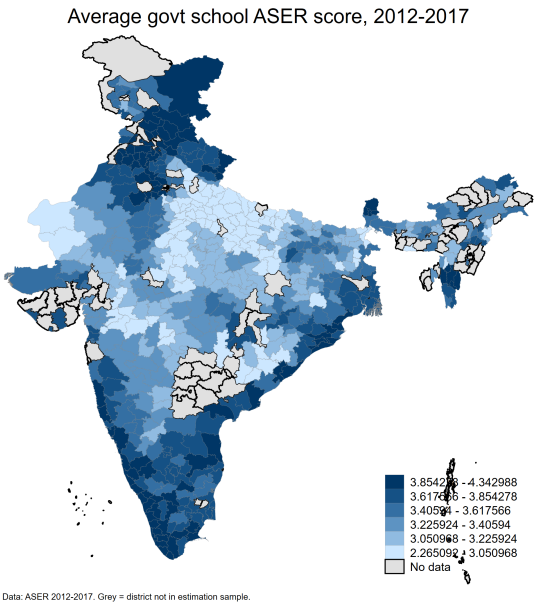
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	Mean	Median	Standard Deviation	Minimum	Maximum	Mean	Median	Standard Deviation	Minimum



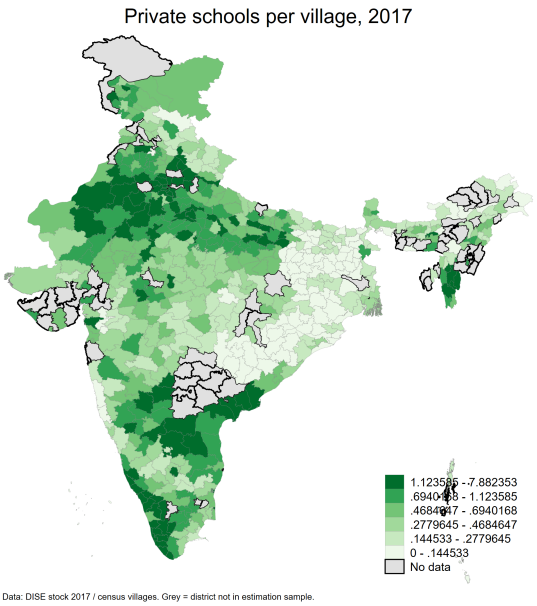
(a) Stock of private schools, 2017



(b) Mean annual entry, 2013–2017

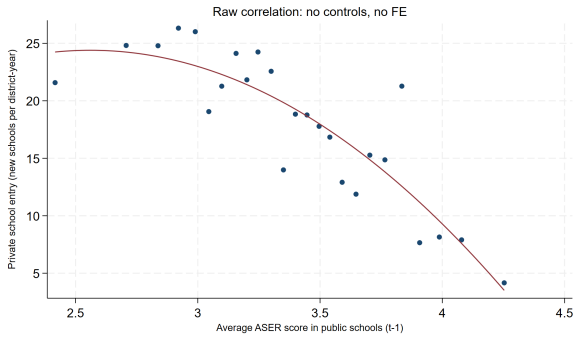


(c) Mean government ASER score, 2012–2017

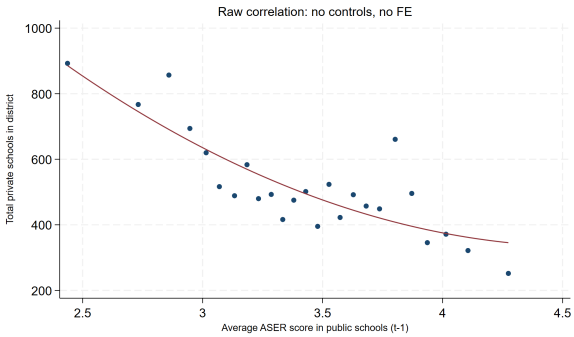


(d) Private schools per census village, 2017

Figure 3: District maps on the rebuilt dataset; 565 districts matched to the 2011 Census shapefile. Panel (d) normalises the stock by the number of census villages and shows that the dispersion in panel (a) is not simply district size.

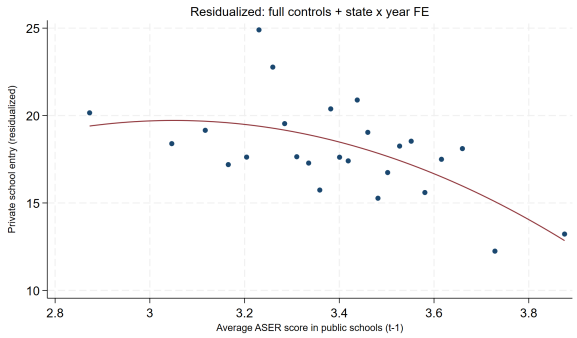


(a) Entry

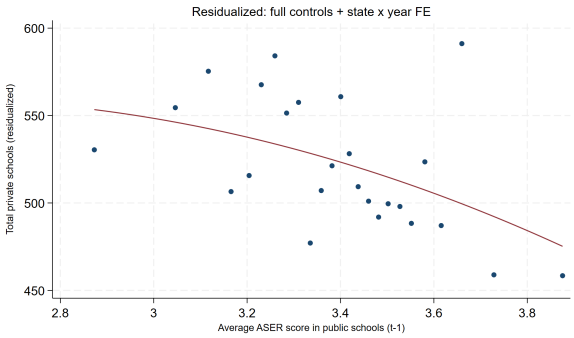


(b) Stock

Figure 4: Unconditional relationship between the lagged government school score and private school entry and stock, on the rebuilt dataset. Twenty-five bins of approximately 80–100 district-years each, with a quadratic fit. Bivariate slopes: entry −11.72 (clustered SE 1.76), $N = 2{,}019$, 540 district clusters; stock −286.6 (clustered SE 50.9), $N = 2{,}514$. Both slope down: private schooling is more prevalent where government schools perform worse.



(a) Entry



(b) Stock

Figure 5: Conditional binscatters, residualised on the main-specification control set and state × year fixed effects. Entry shows a clear interior peak at a score of approximately 3.0–3.1, matching the quadratic estimate of 2.95.

Table 2: Effect of lagged ASER learning levels in public schools on total number of private schools and entry of new private schools

	Total No. of Private Schools			Private School Entry		
	(1)	(2)	(3)	(4)	(5)	(6)
Avg ASER learning in public schools$_{t-1}$	-286.6***	-60.80	854.1**	-11.72***	-5.939***	50.08***
	(50.89)	(47.39)	(428.3)	(1.763)	(2.127)	(16.23)
Avg ASER learning in all schools$^2_{t-1}$			-138.4**			-8.476***
			(60.44)			(2.367)
Average Parental Education$_{t-1}$		35.90*	33.84*		1.158	1.032
		(18.93)	(18.67)		(0.944)	(0.924)
Total no. of public schools$_{t-1}$		0.0451	0.0490		0.00610***	0.00634***
		(0.0529)	(0.0523)		(0.00235)	(0.00231)
Normalized enrollment in public schools$_{t-1}$		-77.69*	-78.10*		4.248**	4.223**
		(43.40)	(43.08)		(1.987)	(1.967)
Infrastructure index in public schools$_{t-1}$		-4.306	-4.546		-2.292	-2.306
		(48.06)	(47.86)		(1.953)	(1.937)
Light Intensity$_{t-1}$		240.4***	239.0***		8.024***	7.939***
		(39.86)	(40.09)		(1.422)	(1.487)
Census infra index		-95.65**	-98.07**		-2.461	-2.609
		(41.17)	(41.88)		(2.746)	(2.752)
Census infra index × Trend		-2.982	-3.597		0.140	0.102
		(3.730)	(3.722)		(0.410)	(0.418)
Log(Population)		313.5***	312.5***		10.75***	10.69***
		(66.97)	(65.91)		(3.220)	(3.183)
Log(Population) x Trend		-29.70***	-30.44***		-1.667***	-1.712***
		(10.42)	(10.32)		(0.563)	(0.561)
% of SC population		109.5	54.34		-0.774	-4.154
		(303.9)	(304.0)		(21.98)	(22.06)
% of SC population × Trend		35.24	34.09		2.926	2.855
		(36.13)	(36.13)		(3.461)	(3.471)
% of ST population		-23.29	-38.82		-14.66*	-15.61*
		(126.8)	(125.2)		(8.766)	(8.616)
% of ST population × Trend		-2.178	-1.733		0.996	1.023
		(12.78)	(12.68)		(1.349)	(1.336)
Constant	1500.8***	-841.8*	-2252.4***	57.69***	16.61	-69.76*
	(181.8)	(446.8)	(862.3)	(6.180)	(21.50)	(38.34)
Observations	2514	1985	1985	2019	1985	1985
R-squared	0.0452	0.753	0.756	0.0432	0.550	0.554
Mean of Dependent Variable	519.0	523.8	523.8	17.88	18.12	18.12
# of Blocks & Villages	NO	YES	YES	NO	YES	YES
Year FE	NO	YES	YES	NO	YES	YES
State FE	NO	YES	YES	NO	YES	YES
State FE × Year FE	NO	YES	YES	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

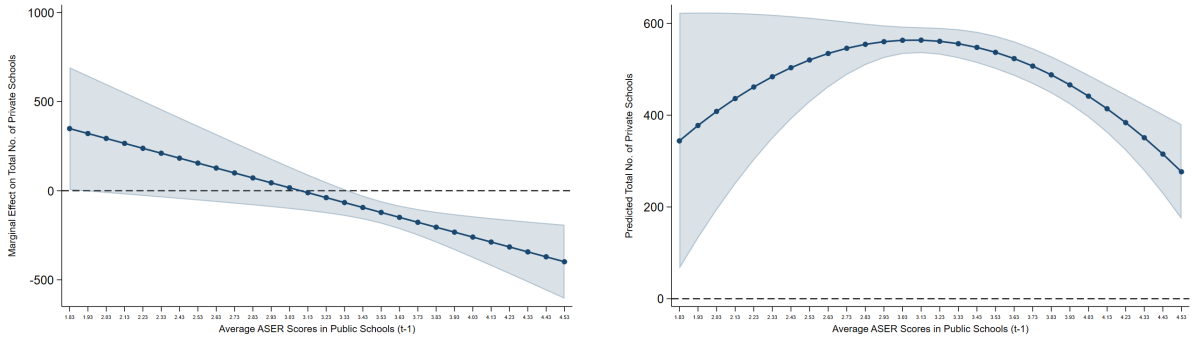


Figure 6: Lagged government school scores and the total number of private schools. Left panel: marginal effect; right panel: predicted margins. Shaded areas are 90 per cent confidence intervals. The outcome is the stock of private schools, so the marginal effect is *net* of exit and should not be read as gross entry; exit is a separate decision and is not modelled here.

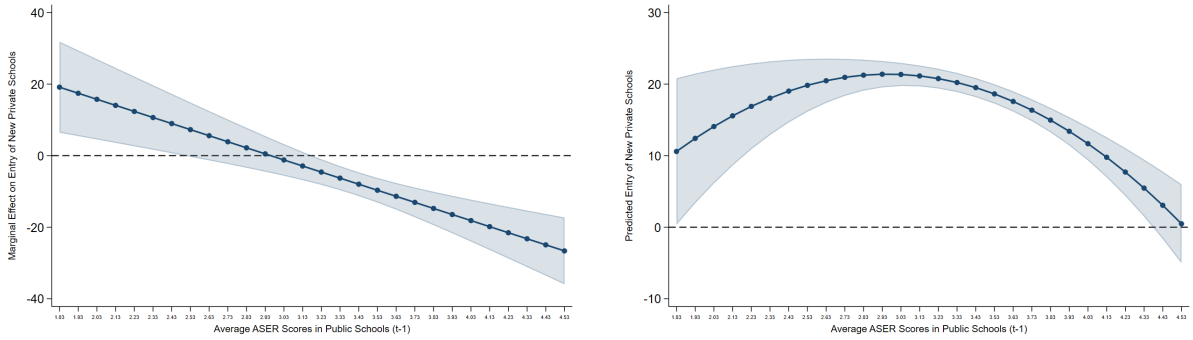


Figure 7: Lagged government school scores and gross entry of new private schools. Shaded areas are 90 per cent confidence intervals.

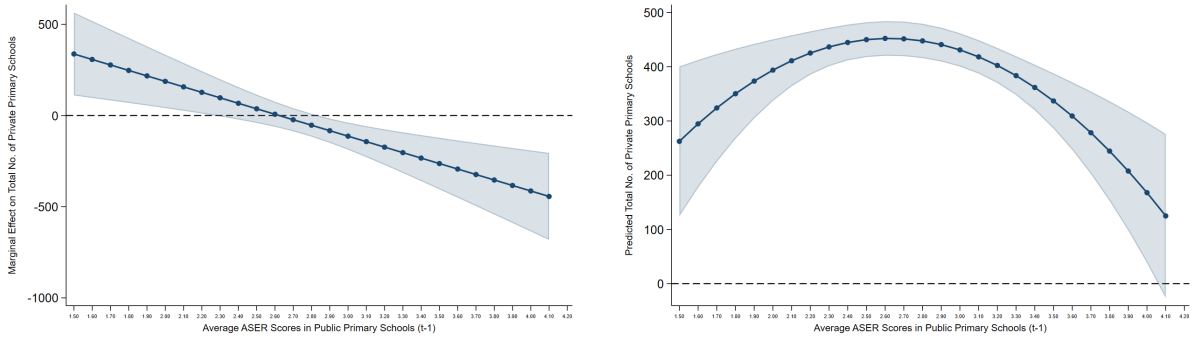


Figure 8: Lagged public primary school scores and the stock of private primary schools. 90 per cent confidence intervals.

Table 3: Entry of private schools: response to the level versus the change in lagged public-school ASER learning

	Private School entry			
	(1)	(2)	(3)	(4)
Avg ASER learning in govt schools$_{t-1}$	-15.06***	-9.731***	61.76***	62.88***
	(2.048)	(2.943)	(17.53)	(17.80)
Avg ASER learning in govt schools$^2_{t-1}$			-10.85***	-10.59***
			(2.565)	(2.596)
ΔAvg ASER learning in govt schools$_{t-1}$	11.59***	8.224***	8.590***	
	(2.641)	(2.494)	(2.495)	
Average Parental Education$_{t-1}$		2.426**	2.285*	2.039*
		(1.229)	(1.201)	(1.171)
Total no. of govt schools$_{t-1}$		0.00791***	0.00826***	0.00807***
		(0.00260)	(0.00254)	(0.00253)
Normalized enrollment in govt schools$_{t-1}$		4.318**	4.209**	4.533**
		(2.098)	(2.065)	(2.068)
Infrastructure index in govt schools$_{t-1}$		-1.561	-1.388	-1.885
		(2.345)	(2.319)	(2.328)
Light Intensity$_{t-1}$		9.386***	9.253***	9.259***
		(1.646)	(1.730)	(1.728)
Census infra index		-5.512	-4.881	-4.867
		(3.402)	(3.422)	(3.379)
Census infra index × Trend		1.038	0.778	0.732
		(0.946)	(0.937)	(0.928)
Log(Population)		1.479	1.561	2.085
		(4.570)	(4.590)	(4.571)
Log(Population) x Trend		0.0214	-0.0726	-0.109
		(1.095)	(1.103)	(1.103)
% of SC population		-37.81	-50.13*	-47.53*
		(28.64)	(28.96)	(28.63)
% of SC population × Trend		11.46	13.37*	13.48*
		(7.138)	(7.245)	(7.190)
% of ST population		-40.19**	-44.13***	-45.20***
		(16.60)	(16.51)	(16.66)
% of ST population × Trend		6.782	7.528*	8.148*
		(4.373)	(4.347)	(4.392)
Constant	71.57***	24.12	-87.06**	-92.84**
	(7.254)	(25.92)	(43.30)	(43.75)
Observations	1447	1422	1422	1422
R-squared	0.0618	0.557	0.563	0.560
Mean of Dependent Variable	19.91	20.18	20.18	20.18
# of Blocks & Villages	NO	YES	YES	YES
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The change regressor is the govt score change from t-2 to t-1. Col. (4): quadratic without the change term, on the col. (3) sample.

Table 4: Effect of lagged ASER learning levels in public primary schools on no. of private primary schools

	Total No. of Private Primary Schools			
	(1)	(2)	(3)	(4)
Avg ASER learning in public primary schools$_{t-1}$	-238.6***	-70.47**	533.1**	787.2***
	(33.28)	(33.16)	(244.4)	(292.6)
Avg ASER learning in public primary schools$^2_{t-1}$			-107.5**	-150.3***
			(42.76)	(52.51)
Average Parental Education$_{t-1}$		17.89	15.33	13.45
		(16.20)	(15.82)	(15.06)
Total no. of public primary schools$_{t-1}$		0.0161	0.0225	0.0255
		(0.0594)	(0.0594)	(0.0597)
Normalized enrollment in public primary schools$_{t-1}$		-59.86*	-58.32*	-59.04*
		(32.85)	(32.61)	(32.87)
Infrastructure index in public primary schools$_{t-1}$		2.544	5.338	71.58
		(31.41)	(31.47)	(58.72)
Light Intensity$_{t-1}$		223.3***	221.5***	226.7***
		(39.23)	(38.87)	(40.97)
Census infra index		-65.47**	-67.16**	-70.69**
		(33.32)	(33.49)	(33.78)
Census infra index × Trend		-3.237	-3.410	-3.872
		(3.408)	(3.390)	(3.633)
Log(Population)		285.6***	284.2***	279.9***
		(58.01)	(57.31)	(57.85)
Log(Population) x Trend		-25.62***	-26.31***	-26.96***
		(9.718)	(9.765)	(10.09)
% of SC population		-4.947	-50.84	-79.91
		(275.0)	(279.7)	(286.1)
% of SC population × Trend		31.65	28.37	23.85
		(34.55)	(34.16)	(33.33)
% of ST population		-52.69	-58.89	-34.99
		(115.9)	(114.9)	(110.5)
% of ST population × Trend		-0.674	-1.101	-3.727
		(11.85)	(11.64)	(11.67)
Avg ASER learning in public post-primary schools$_{t-1}$				-744.0
				(522.4)
Avg ASER learning in public post-primary schools$^2_{t-1}$				89.44
				(63.88)
Infrastructure index in public post-primary schools$_{t-1}$				-106.3
				(81.98)
Constant	1078.8***	-679.7*	-1442.1***	-58.14
	(102.6)	(375.4)	(477.0)	(1078.9)
Observations	2514	1985	1985	1985
R-squared	0.0536	0.734	0.736	0.738
Mean of Dependent Variable	413.6	417.0	417.0	417.0
# of Blocks & Villages	NO	YES	YES	YES
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Effect of lagged ASER learning levels in govt primary schools on entry of private primary schools

	Private Primary School Entry			
	(1)	(2)	(3)	(4)
Avg ASER learning in public primary schools$_{t-1}$	-9.147***	-4.687***	29.68**	33.20**
	(1.439)	(1.695)	(12.33)	(14.94)
Avg ASER learning in public primary schools$^2_{t-1}$			-6.120***	-6.460**
			(2.106)	(2.570)
Average Parental Education$_{t-1}$		0.528	0.382	0.289
		(0.788)	(0.773)	(0.774)
Total no. of public primary schools$_{t-1}$		0.00595**	0.00631**	0.00642**
		(0.00251)	(0.00249)	(0.00251)
Normalized enrollment in public primary schools$_{t-1}$		2.802*	2.890*	2.905*
		(1.587)	(1.579)	(1.577)
Infrastructure index in public primary schools$_{t-1}$		-1.151	-0.992	2.131
		(1.413)	(1.424)	(2.328)
Light Intensity$_{t-1}$		7.233***	7.130***	7.410***
		(1.238)	(1.259)	(1.270)
Census infra index		-2.101	-2.197	-2.301
		(2.564)	(2.591)	(2.599)
Census infra index × Trend		0.0469	0.0370	-0.00759
		(0.411)	(0.419)	(0.419)
Log(Population)		9.679***	9.601***	9.233***
		(3.029)	(3.016)	(3.014)
Log(Population) x Trend		-1.299**	-1.338**	-1.370**
		(0.552)	(0.556)	(0.559)
% of SC population		2.624	0.0114	-1.428
		(20.54)	(20.68)	(20.63)
% of SC population × Trend		1.210	1.023	0.892
		(3.356)	(3.361)	(3.367)
% of ST population		-11.53	-11.88	-11.28
		(8.946)	(9.008)	(9.083)
% of ST population × Trend		0.894	0.870	0.850
		(1.526)	(1.533)	(1.557)
Avg ASER learning in public post-primary schools$_{t-1}$				7.175
				(29.75)
Avg ASER learning in public post-primary schools$^2_{t-1}$				-1.101
				(3.576)
Infrastructure index in public post-primary schools$_{t-1}$				-5.003*
				(2.889)
Constant	39.94***	5.309	-38.10	-44.49
	(4.320)	(16.65)	(25.35)	(58.32)
Observations	2019	1985	1985	1985
R-squared	0.0377	0.516	0.519	0.521
Mean of Dependent Variable	14.73	14.93	14.93	14.93
# of Blocks & Villages	NO	YES	YES	YES
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

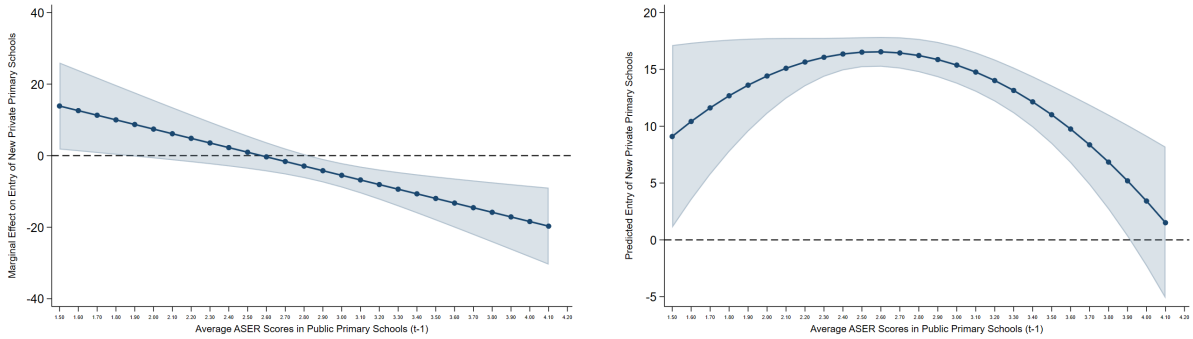


Figure 9: Lagged public primary school scores and entry of private primary schools. 90 per cent confidence intervals.

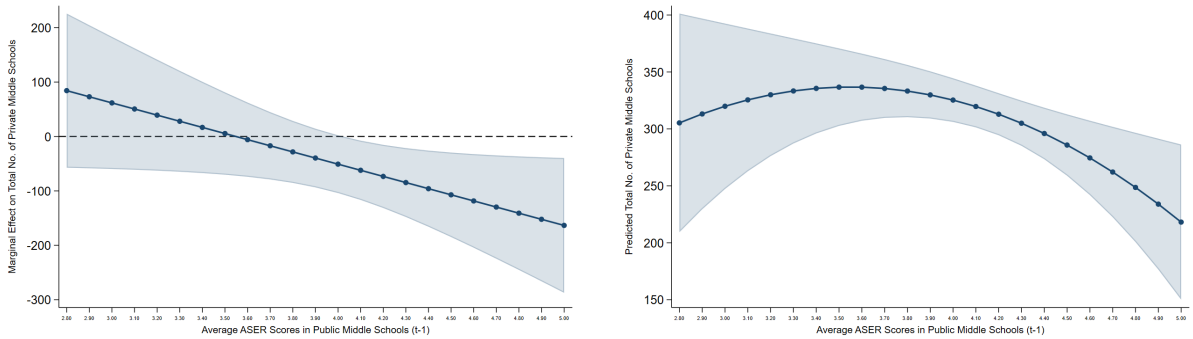


Figure 10: Lagged public middle school scores and the stock of private middle schools. 90 per cent confidence intervals.

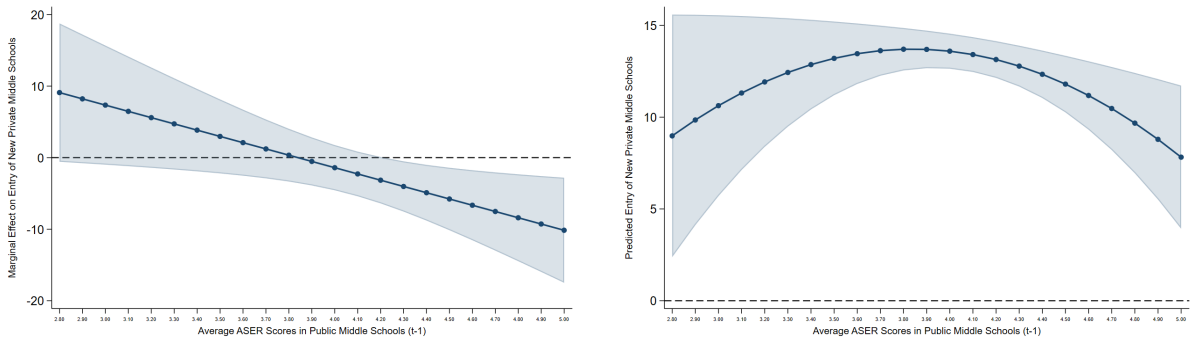


Figure 11: Lagged public middle school scores and entry of private middle schools. 90 per cent confidence intervals.

Table 6: Effect of lagged ASER learning levels in public middle schools on total no. of private middle schools

	Total No. of Private Middle Schools			
	(1)	(2)	(3)	(4)
Avg ASER learning in public middle schools$_{t-1}$	-142.0***	-49.07	291.1	399.3
	(29.68)	(33.20)	(261.4)	(269.9)
Avg ASER learning in public middle schools$^2_{t-1}$			-43.03	-56.30*
			(32.47)	(33.62)
Average Parental Education$_{t-1}$		19.74*	19.55*	21.21*
		(11.85)	(11.81)	(11.41)
Total no. of public middle schools$_{t-1}$		0.462**	0.464**	0.469**
		(0.210)	(0.210)	(0.213)
Infrastructure index in public middle schools$_{t-1}$		2.057	2.244	-16.50
		(27.11)	(27.11)	(29.08)
Normalized enrollment in public middle schools$_{t-1}$		-158.9***	-159.4***	-155.8***
		(45.56)	(45.58)	(44.99)
Light Intensity$_{t-1}$		194.2***	194.3***	192.9***
		(34.53)	(34.55)	(34.68)
Census infra index		-27.75	-28.12	-36.76
		(26.00)	(26.00)	(26.63)
Census infra index × Trend		-4.233	-4.321	-2.363
		(2.957)	(2.955)	(3.123)
Log(Population)		230.0***	228.1***	227.1***
		(49.74)	(49.75)	(51.79)
Log(Population) x Trend		-22.12**	-22.20**	-22.91**
		(9.280)	(9.275)	(9.587)
% of SC population		4.332	6.006	-5.670
		(224.5)	(223.9)	(231.2)
% of SC population × Trend		25.61	23.62	26.43
		(29.32)	(29.30)	(30.70)
% of ST population		-80.34	-85.73	-115.9
		(94.73)	(95.12)	(97.70)
% of ST population × Trend		4.585	5.123	8.176
		(10.06)	(10.11)	(11.59)
Avg ASER learning in public secondary schools$_{t-1}$				-441.7**
				(197.6)
Avg ASER learning in public secondary schools$^2_{t-1}$				49.03**
				(23.81)
Infrastructure index in public secondary schools$_{t-1}$				33.20*
				(18.06)
Constant	889.7***	-700.2*	-1347.2**	-600.1
	(119.6)	(388.2)	(599.2)	(744.9)
Observations	2511	1983	1983	1943
R-squared	0.0165	0.741	0.741	0.743
Mean of Dependent Variable	307.7	313.2	313.2	315.5
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
# of Blocks & Villages	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Effect of lagged ASER learning levels in public middle schools on entry of new private middle schools

	Private Middle School Entry			
	(1)	(2)	(3)	(4)
Avg ASER learning in public middle schools$_{t-1}$	-10.48***	-1.249	29.99	33.57*
	(1.712)	(1.844)	(18.24)	(17.89)
Avg ASER learning in public middle schools$^2_{t-1}$			-3.951*	-4.373**
			(2.205)	(2.184)
Average Parental Education$_{t-1}$		-0.141	-0.159	-0.202
		(0.755)	(0.753)	(0.699)
Total no. of public middle schools$_{t-1}$		0.0119**	0.0120**	0.0120**
		(0.00526)	(0.00525)	(0.00533)
Infrastructure index in public middle schools$_{t-1}$		-3.431**	-3.414**	-4.629**
		(1.494)	(1.488)	(1.971)
Normalized enrollment in public middle schools$_{t-1}$		-1.183	-1.233	-1.118
		(1.409)	(1.405)	(1.386)
Light Intensity$_{t-1}$		6.323***	6.335***	6.288***
		(0.865)	(0.862)	(0.868)
Census infra index		-2.251	-2.285	-2.942
		(2.267)	(2.288)	(2.256)
Census infra index × Trend		-0.0581	-0.0661	0.0493
		(0.370)	(0.375)	(0.377)
Log(Population)		6.331**	6.153**	5.865**
		(2.743)	(2.729)	(2.749)
Log(Population) x Trend		-0.311	-0.318	-0.314
		(0.399)	(0.400)	(0.409)
% of SC population		-19.93	-19.77	-21.22
		(17.46)	(17.60)	(17.71)
% of SC population × Trend		4.890	4.707	5.148*
		(3.069)	(3.092)	(3.113)
% of ST population		-6.289	-6.784	-4.765
		(8.107)	(8.101)	(8.495)
% of ST population × Trend		0.728	0.777	0.349
		(1.274)	(1.276)	(1.380)
Avg ASER learning in public secondary schools$_{t-1}$				-7.306
				(12.75)
Avg ASER learning in public secondary schools$^2_{t-1}$				0.779
				(1.571)
Infrastructure index in public secondary schools$_{t-1}$				2.091
				(1.627)
Constant	55.31***	-1.965	-61.37	-51.93
	(7.170)	(17.21)	(40.12)	(48.12)
Observations	2017	1983	1983	1943
R-squared	0.0358	0.529	0.530	0.532
Mean of Dependent Variable	12.85	13.04	13.04	12.95
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
# of Blocks & Villages	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Effect of lagged ASER learning levels in public secondary schools on entry of new private secondary schools

	Private secondary school entry		
	(1)	(2)	(3)
Avg ASER learning in public secondary schools$_{t-1}$	-1.920	1.307	-14.00
	(1.289)	(1.906)	(17.32)
Avg ASER learning in public secondary schools$^2_{t-1}$			1.811
			(2.224)
Average Parental Education$_{t-1}$		0.0603	0.0413
		(0.408)	(0.418)
Total no. of public secondary schools$_{t-1}$		0.0157*	0.0157*
		(0.00832)	(0.00832)
Infrastructure index in public secondary schools$_{t-1}$		-0.693	-0.727
		(1.721)	(1.750)
Normalized enrollment in public secondary schools$_{t-1}$		-2.148**	-2.117**
		(1.049)	(1.056)
Light Intensity$_{t-1}$		4.076***	4.069***
		(0.628)	(0.627)
Census infra index		-3.894*	-4.005*
		(2.146)	(2.205)
Census infra index × Trend		0.540	0.568
		(0.385)	(0.399)
Log(Population)		3.313	3.422
		(3.385)	(3.461)
Log(Population) x Trend		-0.305	-0.308
		(0.489)	(0.490)
% of SC population		4.387	4.970
		(16.34)	(16.38)
% of SC population × Trend		0.767	0.694
		(3.002)	(3.002)
% of ST population		1.719	1.399
		(5.887)	(6.060)
% of ST population × Trend		-0.149	-0.0683
		(1.097)	(1.149)
Constant	15.28***	-11.65	20.01
	(5.667)	(12.77)	(33.43)
Observations	2014	1942	1942
R-squared	0.00119	0.338	0.339
Mean of Dependent Variable	6.685	6.819	6.819
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

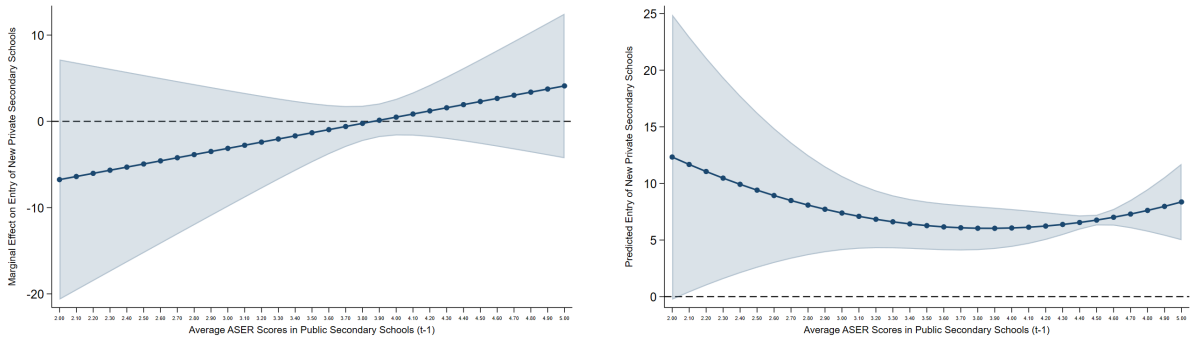


Figure 12: Lagged public secondary school scores and entry of private secondary schools. No relationship is detected at this level. 90 per cent confidence intervals.

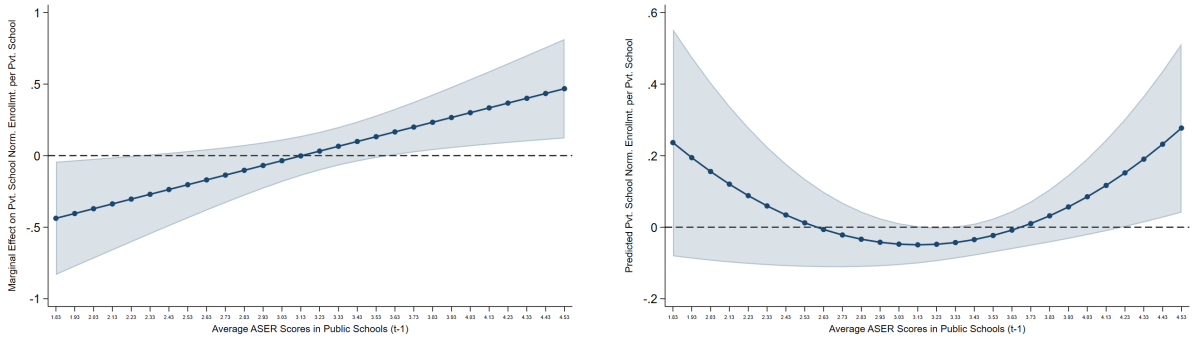


Figure 13: Lagged government school scores and enrolment per operating private school (within-year z-score). U-shaped, with a minimum at a score of approximately 3.13. 90 per cent confidence intervals.

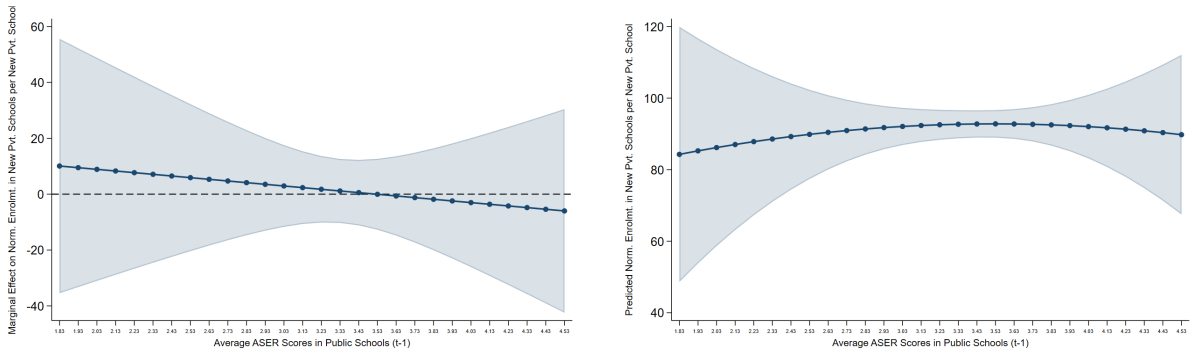


Figure 14: Lagged government school scores and enrolment per new private school (raw pupils; conditions on districts with positive entry). Note the scaling differs from Figure 13, which is a within-year z-score. 90 per cent confidence intervals.

Table 9: Effect of lagged ASER learning levels in public schools on ratio of Private school (normalized) enrollment to total no. of private schools

	Private School Normalized Enrollment/Total Private Schools		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.166*	0.0585	-1.050**
	(0.0857)	(0.0820)	(0.518)
Avg ASER learning in public schools$^2_{t-1}$			0.168**
			(0.0781)
Average Parental Education$_{t-1}$		0.00696	0.00946
		(0.0294)	(0.0294)
Total no. of public schools$_{t-1}$		-0.000148**	-0.000153***
		(0.0000584)	(0.0000583)
Infrastructure index in public schools$_{t-1}$		0.0411	0.0414
		(0.0702)	(0.0700)
Normalized enrollment in public schools$_{t-1}$		-0.00193	-0.00143
		(0.0544)	(0.0543)
Light Intensity$_{t-1}$		0.131***	0.132**
		(0.0504)	(0.0520)
Census infra index		0.0109	0.0138
		(0.0831)	(0.0831)
Census infra index × Trend		0.00839	0.00914
		(0.00934)	(0.00939)
Log(Population)		0.552***	0.553***
		(0.105)	(0.106)
Log(Population) x Trend		-0.0130	-0.0121
		(0.0100)	(0.00998)
% of SC population		-0.460	-0.393
		(0.702)	(0.701)
% of SC population × Trend		-0.134	-0.133
		(0.0983)	(0.0970)
% of ST population		-0.305	-0.286
		(0.296)	(0.297)
% of ST population × Trend		0.0353	0.0348
		(0.0355)	(0.0359)
Constant	-0.593**	-3.530***	-1.821
	(0.283)	(0.907)	(1.123)
Observations	2514	1985	1985
R-squared	0.00652	0.701	0.702
Mean of Dependent Variable	-0.0232	-0.00511	-0.00511
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Effect of lagged ASER learning levels in public schools on ratio of (normalized) enrolment in new private schools to number of new private schools

	Normalized Private School Enrollment/New Private Schools		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	-33.12***	1.297	20.95
	(6.549)	(6.986)	(59.54)
Avg ASER learning in public schools$^2_{t-1}$			-2.978
			(8.859)
Average Parental Education$_{t-1}$		3.226	3.147
		(2.746)	(2.786)
Total no. of public schools$_{t-1}$		0.00104	0.00116
		(0.00721)	(0.00726)
Infrastructure index in public schools$_{t-1}$		-3.297	-3.275
		(5.153)	(5.163)
Normalized enrollment in public schools$_{t-1}$		-2.101	-2.111
		(5.242)	(5.242)
Light Intensity$_{t-1}$		4.020**	4.012**
		(1.708)	(1.698)
Census infra index		-4.496	-4.567
		(12.64)	(12.62)
Census infra index × Trend		0.354	0.334
		(2.324)	(2.326)
Log(Population)		35.71**	35.51**
		(14.03)	(14.04)
Log(Population) x Trend		-3.326	-3.332
		(2.648)	(2.649)
% of SC population		82.13	80.89
		(129.4)	(130.0)
% of SC population × Trend		-17.39	-17.45
		(23.85)	(23.85)
% of ST population		-28.25	-28.68
		(39.13)	(39.54)
% of ST population × Trend		4.124	4.138
		(7.222)	(7.236)
Constant	203.3***	-62.59	-91.60
	(23.30)	(59.86)	(111.4)
Observations	2226	1730	1730
R-squared	0.0337	0.463	0.463
Mean of Dependent Variable	94.75	92.18	92.18
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11: Effect of lagged ASER learning levels in public schools on ratio of enrolment in private schools to enrolment in public schools

	Private School Enrollment/Public School Enrollment		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.0474	0.142***	-0.0685
	(0.0548)	(0.0492)	(0.349)
Avg ASER learning in public schools$^2_{t-1}$			0.0318
			(0.0547)
Average Parental Education$_{t-1}$		0.0675***	0.0679***
		(0.0201)	(0.0200)
Total no. of public schools$_{t-1}$		-0.000257***	-0.000257***
		(0.0000473)	(0.0000471)
Infrastructure index in public schools$_{t-1}$		0.123***	0.123***
		(0.0436)	(0.0435)
Light Intensity$_{t-1}$		0.236***	0.237***
		(0.0249)	(0.0250)
Census infra index		-0.110*	-0.109*
		(0.0563)	(0.0564)
Census infra index × Trend		0.0113*	0.0114*
		(0.00672)	(0.00667)
Log(Population)		0.426***	0.427***
		(0.0786)	(0.0788)
Log(Population) x Trend		-0.0171**	-0.0169**
		(0.00773)	(0.00777)
% of SC population		-0.0929	-0.0802
		(0.523)	(0.530)
% of SC population × Trend		-0.0349	-0.0346
		(0.0693)	(0.0691)
% of ST population		0.0198	0.0234
		(0.205)	(0.206)
% of ST population × Trend		0.0157	0.0156
		(0.0262)	(0.0263)
Constant	0.570***	-2.741***	-2.417***
	(0.207)	(0.478)	(0.751)
Observations	2514	1985	1985
R-squared	0.0252	0.720	0.721
Mean of Dependent Variable	0.588	0.601	0.601
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

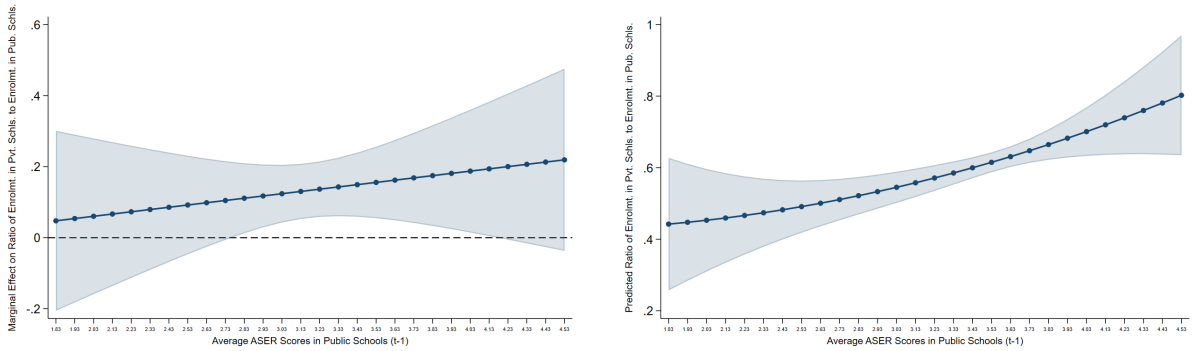


Figure 15: Lagged government school scores and the ratio of private to government enrolment. The conditional slope is positive and linear, with no detectable curvature. 90 per cent confidence intervals.

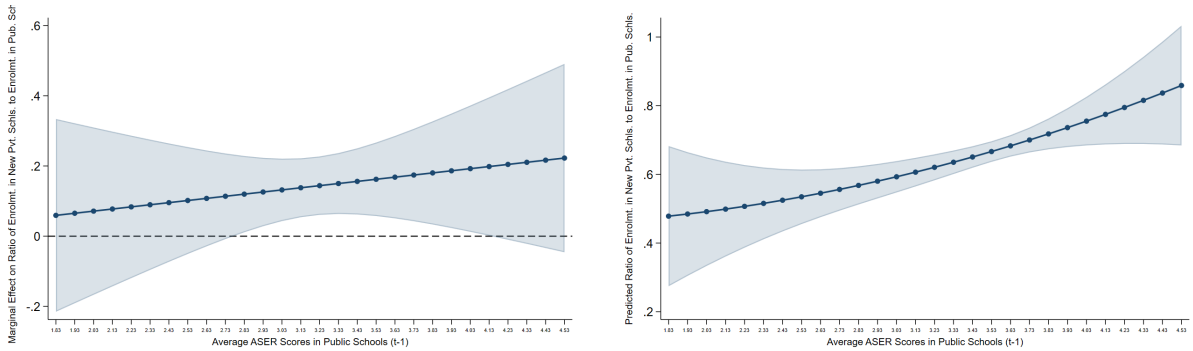


Figure 16: Lagged government school scores and the ratio of enrolment in new private schools to enrolment in government schools. 90 per cent confidence intervals.

Main specification: size of the private sector

By level offered

Intensive margin: size of private schools

Private relative to government enrolment

Private school quality

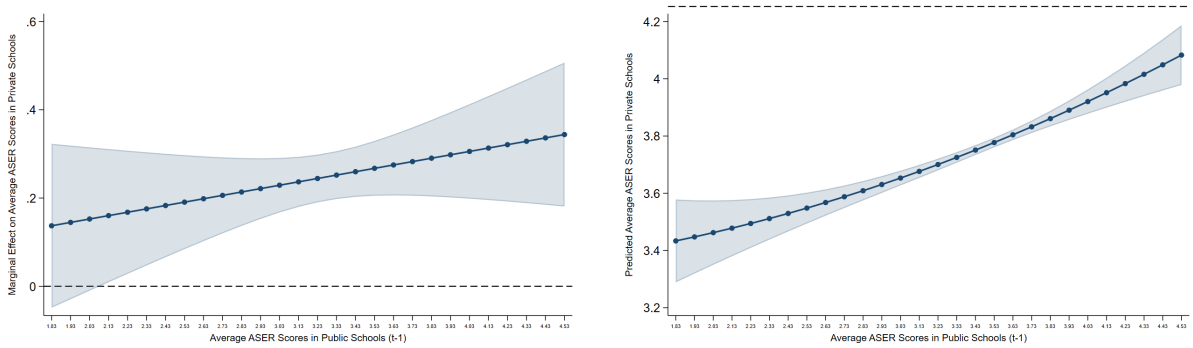


Figure 17: Lagged government school scores and average private school ASER scores. Monotone and stable at approximately +0.25 per score point, with no curvature. 90 per cent confidence intervals.

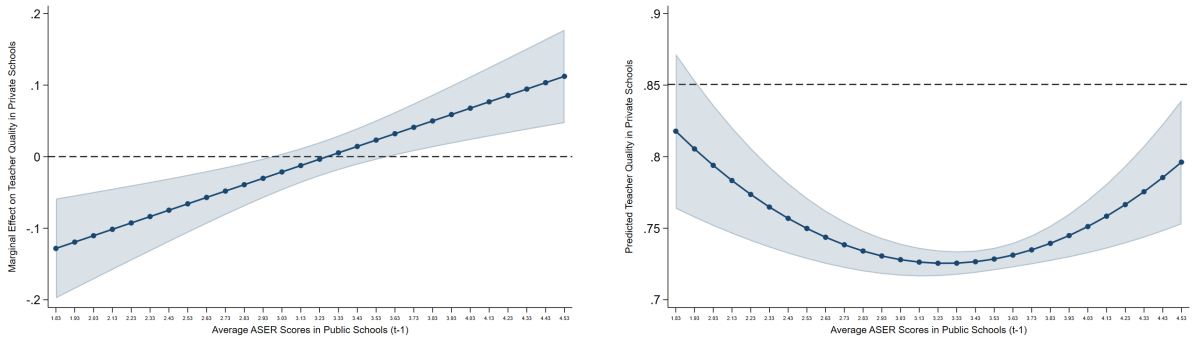


Figure 18: Lagged government school scores and the share of private school teachers holding a graduate degree. U-shaped, with a minimum at approximately 3.26. This margin does not pass the leads placebo (Table 19) and should be read as a persistent district-level association. 90 per cent confidence intervals.

Dynamics and placebo tests

Functional form

Heterogeneity

Ordinal robustness

Appendix: alternative measure of public school quality

Table 12: Effect of lagged ASER learning levels in public schools on average learning in private schools

	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.254***	0.250***	-0.00263
	(0.0187)	(0.0328)	(0.247)
Avg ASER learning in public schools$^2_{t-1}$			0.0383
			(0.0374)
Average Parental Education$_{t-1}$		0.0289**	0.0299**
		(0.0142)	(0.0141)
Total no. of public schools$_{t-1}$		0.0000516**	0.0000502**
		(0.0000239)	(0.0000239)
Infrastructure index in public schools$_{t-1}$		0.0603**	0.0593**
		(0.0281)	(0.0284)
Normalized enrollment in public schools$_{t-1}$		-0.0387*	-0.0385*
		(0.0215)	(0.0214)
Light Intensity$_{t-1}$		0.0165**	0.0171**
		(0.00826)	(0.00844)
Census infra index		0.134	0.131
		(0.110)	(0.110)
Census infra index × Trend		-0.0477	-0.0465
		(0.0293)	(0.0293)
Log(Population)		-0.0799	-0.0782
		(0.101)	(0.101)
Log(Population) x Trend		-0.000890	-0.000866
		(0.0269)	(0.0269)
% of SC population		0.985	1.038
		(0.826)	(0.834)
% of SC population × Trend		-0.166	-0.177
		(0.234)	(0.236)
% of ST population		-0.532	-0.515
		(0.415)	(0.414)
% of ST population × Trend		0.0912	0.0874
		(0.116)	(0.116)
Constant	2.868***	2.976***	3.360***
	(0.0641)	(0.301)	(0.493)
Observations	1460	949	949
R-squared	0.143	0.473	0.473
Mean of Dependent Variable	3.744	3.748	3.748
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13: Effect of lagged ASER learning levels in public schools on teacher quality in private schools

	Private School Teacher Quality		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.0223	0.00353	-0.291***
	(0.0138)	(0.0149)	(0.0929)
Avg ASER learning in public schools$^2_{t-1}$			0.0446***
			(0.0142)
Average Parental Education$_{t-1}$		0.0135**	0.0142**
		(0.00592)	(0.00574)
Total no. of public schools$_{t-1}$		-0.00000852	-0.00000976
		(0.0000110)	(0.0000109)
Infrastructure index in public schools$_{t-1}$		-0.00287	-0.00279
		(0.0146)	(0.0143)
Normalized enrollment in public schools$_{t-1}$		-0.0399***	-0.0398***
		(0.0123)	(0.0123)
Light Intensity$_{t-1}$		0.00261	0.00305
		(0.00498)	(0.00525)
Census infra index		-0.00408	-0.00330
		(0.0187)	(0.0186)
Census infra index × Trend		0.000593	0.000791
		(0.00270)	(0.00270)
Log(Population)		0.0712***	0.0715***
		(0.0251)	(0.0249)
Log(Population) x Trend		-0.00409	-0.00385
		(0.00264)	(0.00262)
% of SC population		-0.103	-0.0855
		(0.147)	(0.147)
% of SC population × Trend		0.00752	0.00790
		(0.0244)	(0.0245)
% of ST population		-0.0474	-0.0424
		(0.0589)	(0.0593)
% of ST population × Trend		0.000315	0.000172
		(0.00871)	(0.00875)
Constant	0.647***	0.244*	0.698***
	(0.0477)	(0.147)	(0.202)
Observations	2514	1985	1985
R-squared	0.00286	0.787	0.789
Mean of Dependent Variable	0.723	0.735	0.735
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Effect of lagged ASER learning levels in public schools on Teacher Quality in New Private Schools

	Teacher Quality in New Private Schools		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.0174	0.00236	-0.351*
	(0.0153)	(0.0247)	(0.184)
Avg ASER learning in public schools$^2_{t-1}$			0.0536*
			(0.0283)
Average Parental Education$_{t-1}$		-0.00154	-0.000173
		(0.00955)	(0.00924)
Total no. of public schools$_{t-1}$		0.0000239	0.0000216
		(0.0000150)	(0.0000149)
Infrastructure index in public schools$_{t-1}$		0.0189	0.0181
		(0.0225)	(0.0222)
Normalized enrollment in public schools$_{t-1}$		-0.0523***	-0.0522***
		(0.0150)	(0.0149)
Light Intensity$_{t-1}$		0.00214	0.00248
		(0.00542)	(0.00550)
Census infra index		-0.000570	0.00112
		(0.0408)	(0.0403)
Census infra index × Trend		-0.00394	-0.00370
		(0.00777)	(0.00777)
Log(Population)		0.0358	0.0396
		(0.0361)	(0.0359)
Log(Population) x Trend		-0.00584	-0.00582
		(0.00651)	(0.00651)
% of SC population		0.0122	0.0426
		(0.310)	(0.305)
% of SC population × Trend		-0.0504	-0.0515
		(0.0647)	(0.0644)
% of ST population		-0.0185	-0.00810
		(0.124)	(0.123)
% of ST population × Trend		-0.00991	-0.0107
		(0.0243)	(0.0242)
Constant	0.670***	0.662***	1.186***
	(0.0536)	(0.229)	(0.362)
Observations	2178	1685	1685
R-squared	0.00106	0.536	0.538
Mean of Dependent Variable	0.730	0.742	0.742
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

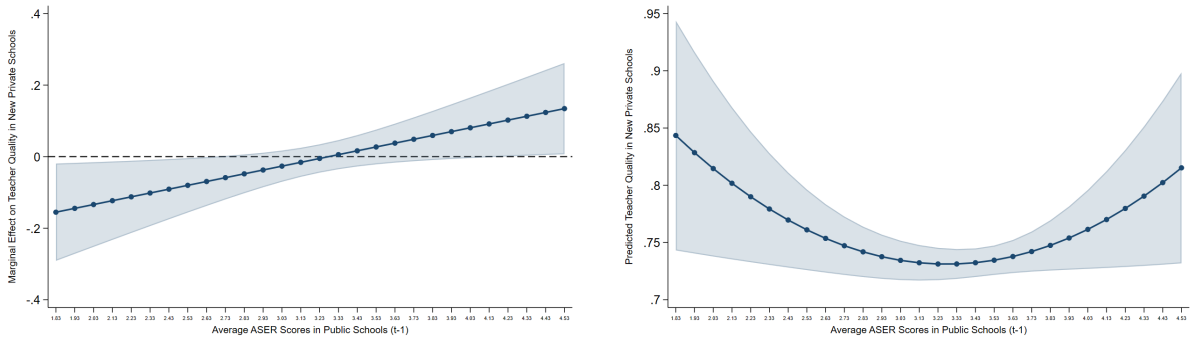


Figure 19: Lagged government school scores and teacher qualifications in new private schools. 90 per cent confidence intervals.

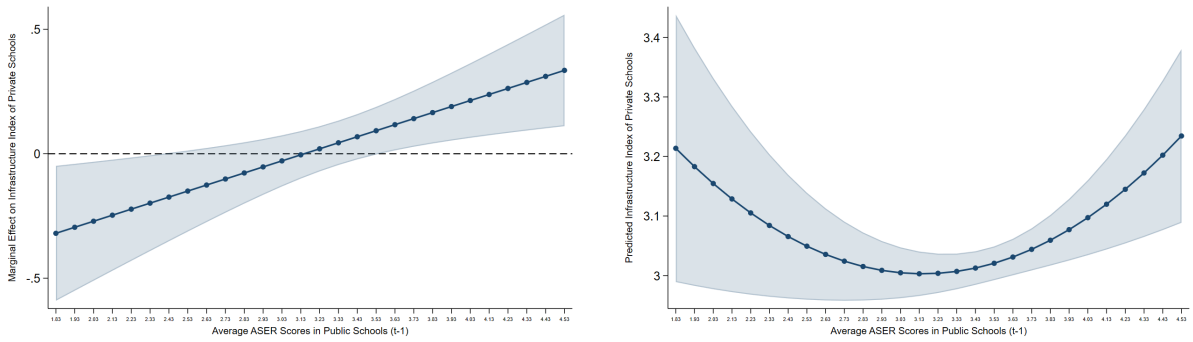


Figure 20: Lagged government school scores and the private school infrastructure index (quintile index from a 13-input principal components analysis). U-shaped, with a minimum at approximately 3.14. 90 per cent confidence intervals.

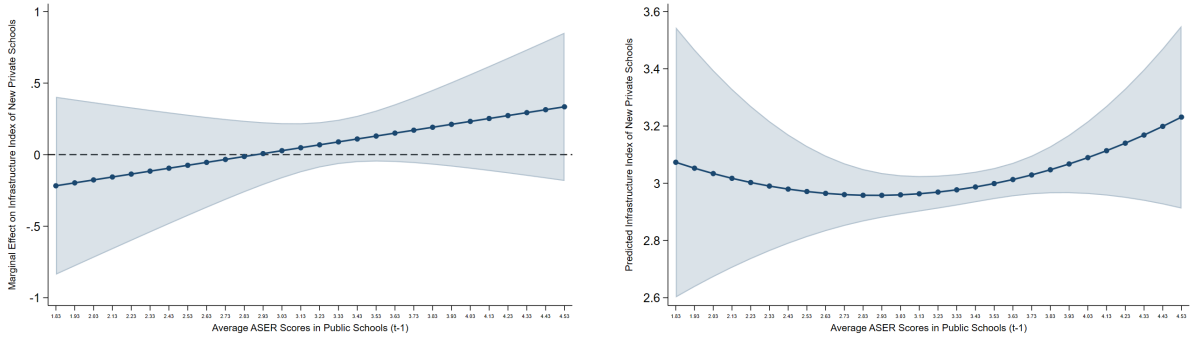


Figure 21: Lagged government school scores and the infrastructure index of new private schools. 90 per cent confidence intervals.

Table 15: Effect of lagged ASER learning levels in public schools on infrastructure index of all private schools

	Private School Infrastructure Index		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.641***	0.0391	-0.762**
	(0.0658)	(0.0544)	(0.349)
Avg ASER learning in public schools$^2_{t-1}$			0.121**
			(0.0520)
Average Parental Education$_{t-1}$		-0.0344	-0.0326
		(0.0212)	(0.0211)
Total no. of public schools$_{t-1}$		0.0000328	0.0000294
		(0.0000340)	(0.0000337)
Infrastructure index in public schools$_{t-1}$		0.336***	0.336***
		(0.0487)	(0.0491)
Normalized enrollment in public schools$_{t-1}$		-0.188***	-0.188***
		(0.0385)	(0.0385)
Light Intensity$_{t-1}$		0.0770***	0.0782***
		(0.0239)	(0.0250)
Census infra index		-0.0885	-0.0863
		(0.0593)	(0.0594)
Census infra index × Trend		0.00361	0.00415
		(0.00799)	(0.00801)
Log(Population)		0.314***	0.315***
		(0.0810)	(0.0805)
Log(Population) x Trend		-0.0195**	-0.0189**
		(0.00904)	(0.00894)
% of SC population		-0.115	-0.0665
		(0.481)	(0.483)
% of SC population × Trend		-0.00798	-0.00697
		(0.0771)	(0.0769)
% of ST population		0.0479	0.0615
		(0.211)	(0.211)
% of ST population × Trend		-0.0120	-0.0124
		(0.0320)	(0.0321)
Constant	1.050***	0.777	2.013***
	(0.239)	(0.555)	(0.715)
Observations	2514	1985	1985
R-squared	0.195	0.837	0.838
Mean of Dependent Variable	2.990	3.034	3.034
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16: Effect of lagged ASER learning levels in public schools on the infrastructure index of new private schools

	Infrastructure Index of New Private Schools		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.508***	0.0813	-0.591
	(0.0757)	(0.0937)	(0.822)
Avg ASER learning in public schools$^2_{t-1}$			0.102
			(0.124)
Average Parental Education$_{t-1}$		0.00711	0.00975
		(0.0362)	(0.0366)
Total no. of public schools$_{t-1}$		0.0000848	0.0000801
		(0.0000638)	(0.0000643)
Infrastructure index in public schools$_{t-1}$		0.410***	0.409***
		(0.0900)	(0.0904)
Normalized enrollment in public schools$_{t-1}$		-0.165**	-0.165**
		(0.0671)	(0.0672)
Light Intensity$_{t-1}$		0.0536**	0.0538**
		(0.0230)	(0.0231)
Census infra index		-0.124	-0.121
		(0.168)	(0.169)
Census infra index × Trend		-0.00422	-0.00348
		(0.0345)	(0.0345)
Log(Population)		0.304*	0.312*
		(0.178)	(0.179)
Log(Population) x Trend		-0.0516	-0.0512
		(0.0347)	(0.0348)
% of SC population		1.024	1.116
		(1.447)	(1.452)
% of SC population × Trend		-0.232	-0.243
		(0.312)	(0.312)
% of ST population		-0.429	-0.393
		(0.616)	(0.616)
% of ST population × Trend		-0.0212	-0.0266
		(0.125)	(0.124)
Constant	1.445***	1.128	2.107
	(0.268)	(0.936)	(1.575)
Observations	1994	1515	1515
R-squared	0.0650	0.523	0.524
Mean of Dependent Variable	2.991	3.000	3.000
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 17: Distributed lag: private school entry and the first two lags of government school quality

	Private school entry		
	(1)	(2)	(3)
Avg ASER learning in govt schools$_{t-1}$	62.88***		82.72***
	(17.80)		(17.80)
Avg ASER learning in govt schools$^2_{t-1}$	-10.59***		-12.72***
	(2.596)		(2.568)
Avg ASER learning in govt schools$_{t-2}$		21.62	-28.23
		(19.62)	(21.04)
Avg ASER learning in govt schools$^2_{t-2}$		-4.615	2.937
		(2.829)	(2.997)
Observations	1422	1422	1422
Districts	499	499	499
R-squared	0.560	0.558	0.564
Mean of Dependent Variable	20.18	20.18	20.18
Full control set	YES	YES	YES
State FE × Year FE	YES	YES	YES

All columns estimated on the identical sample of district-years for which both lags are observed.

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 18: Lead placebo: future government school quality does not predict current private entry

	Private school entry			
	(1)	(2)	(3)	(4)
Avg ASER learning in govt schools$_{t-1}$	49.27***	49.04***	35.35**	
	(16.84)	(16.84)	(16.33)	
Avg ASER learning in govt schools$^2_{t-1}$	-8.149***	-8.143***	-6.070**	
	(2.440)	(2.440)	(2.357)	
Avg ASER learning in govt schools$_{t+1}$		0.327	24.96	49.40***
		(2.219)	(17.47)	(18.35)
Avg ASER learning in govt schools$^2_{t+1}$			-3.719	-7.843***
			(2.541)	(2.673)
Observations	1479	1479	1479	1479
Districts	528	528	528	528
R-squared	0.538	0.538	0.538	0.535
Mean of Dependent Variable	17.43	17.43	17.43	17.43
Full control set	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES

All columns estimated on the identical sample of district-years for which both the lag and the lead are observed.

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

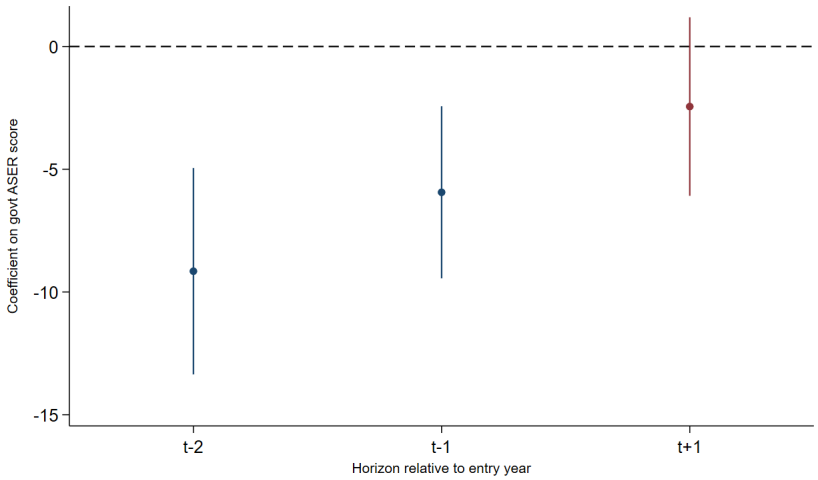


Figure 22: Linear-in-score coefficient on private school entry at each horizon, each estimated on its own maximal sample: $t-2$, -9.15 (2.55); $t-1$, -5.94 (2.13); $t+1$, -2.45 (2.21). The signal is present before entry and absent after it. 90 per cent confidence intervals.

Table 19: Lead placebo across the quantity and quality margins

	private_sch	avg_scoreprivate	prop_pvttch_gradabove_wue	pvtschinfraindex_wue
Govt $ASER_{t-1}$	588.5*	0.141	-0.222***	-0.425
	(309.5)	(0.362)	(0.0796)	(0.305)
Govt $ASER^2_{t-1}$	-99.17**	-0.00600	0.0328***	0.0758
	(44.65)	(0.0537)	(0.0121)	(0.0465)
Govt $ASER_{t+1}$	497.7	-0.376	-0.163**	-0.453
	(374.6)	(0.454)	(0.0817)	(0.333)
Govt $ASER^2_{t+1}$	-74.36	0.0910	0.0278**	0.0644
	(53.94)	(0.0663)	(0.0120)	(0.0495)
Observations	1479	468	1479	1479
Districts	528	468	528	528
R-squared	0.751	0.459	0.796	0.834
Mean of Dependent Variable	523.7	3.740	0.740	3.014
Full control set	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

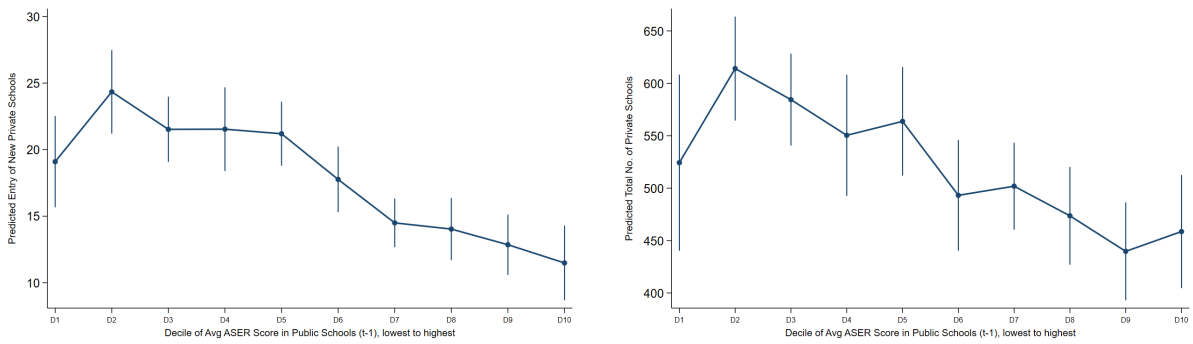
Table 20: Two lags and one lead of government school quality, estimated jointly

	Private school entry	
	(1)	(2)
Avg ASER learning in govt schools$_{t-1}$	-0.956	66.41***
	(3.513)	(23.33)
Avg ASER learning in govt schools$^2_{t-1}$		-10.19***
		(3.294)
Avg ASER learning in govt schools$_{t-2}$	-7.305**	-35.64
	(3.391)	(27.27)
Avg ASER learning in govt schools$^2_{t-2}$		4.155
		(3.824)
Avg ASER learning in govt schools$_{t+1}$	1.001	38.68*
	(3.108)	(22.28)
Avg ASER learning in govt schools$^2_{t+1}$		-5.737*
		(3.180)
Observations	937	937
Districts	495	495
R-squared	0.537	0.545
Mean of Dependent Variable	19.87	19.87
Full control set	YES	YES
State FE × Year FE	YES	YES

The ASER calendar (waves in 2012, 2013, 2014, 2016) admits only one year in which all three horizons are observed.

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$



(a) Entry

(b) Stock

Figure 23: Predicted margins by decile of the lagged government school score, replacing the quadratic with bin dummies. Quintile versions are available as a coarser alternative.

Table 21: Lags and leads defined on the ASER wave calendar

	Private school entry		
	(1)	(2)	(3)
Avg ASER learning in govt schools$_{\text{prev. wave}}$	28.19	16.22	71.75
	(19.63)	(22.73)	(48.74)
Avg ASER learning in govt schools$^2_{\text{prev. wave}}$	-5.769**	-3.775	-11.35
	(2.926)	(3.357)	(6.963)
Avg ASER learning in govt schools$_{\text{2 waves back}}$			-23.53
			(44.48)
Avg ASER learning in govt schools$^2_{\text{2 waves back}}$			1.959
			(6.353)
Avg ASER learning in govt schools$_{\text{next wave}}$		22.50	
		(21.44)	
Avg ASER learning in govt schools$^2_{\text{next wave}}$		-3.719	
		(3.088)	
Observations	956	956	470
Districts	499	499	470
R-squared	0.607	0.608	0.603
Mean of Dependent Variable	19.62	19.62	21.11
Full control set	YES	YES	YES
State FE × Year FE	YES	YES	YES

ASER waves: 2012, 2013, 2014, 2016. Intervals between waves are unequal.

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

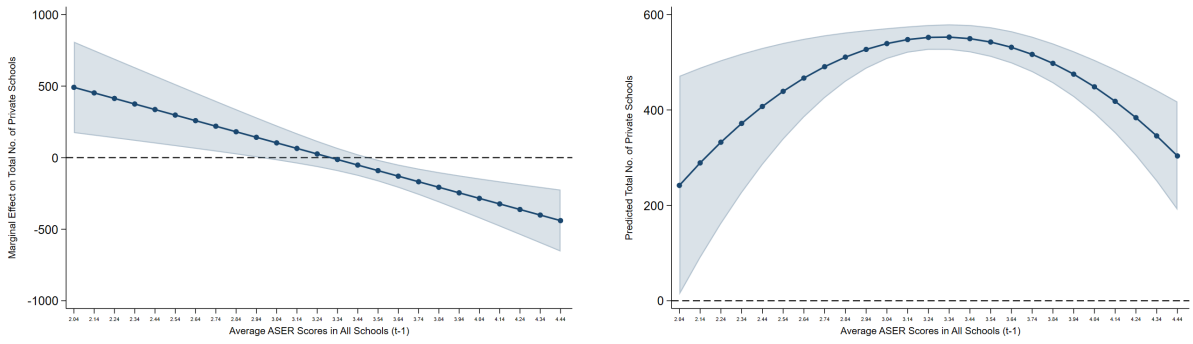


Figure 24: Lagged scores in all schools and the total number of private schools. 90 per cent confidence intervals.

Table 22: Functional-form robustness: entry of private schools

	Private School Entry			
	Quadratic	Quintile bins	Decile bins	Spline
	(1)	(2)	(3)	(4)
Avg ASER learning in govt schools$_{t-1}$	50.08***			
	(16.23)			
Avg ASER learning in govt schools$^2_{t-1}$	-8.476***			
	(2.367)			
Score quintile (base: Q1, lowest)				
Q2		-0.706		
		(1.792)		
Q3		-2.896		
		(1.963)		
Q4		-8.030***		
		(2.014)		
Q5		-9.864***		
		(2.317)		
Score decile (base: D1, lowest)				
D2			5.248**	
			(2.535)	
D3			2.431	
			(2.254)	
D4			2.442	
			(3.065)	
D5			2.099	
			(2.575)	
D6			-1.331	
			(2.827)	
D7			-4.597*	
			(2.595)	
D8			-5.065*	
			(2.737)	
D9			-6.240**	
			(2.887)	
D10			-7.609**	
			(3.102)	
Spline slope 1: below 3.04				10.16**
				(5.025)
Spline slope 2: 3.04 to 3.31				-7.800
				(8.383)
Spline slope 3: 3.31 to 3.55				-18.71**
				(8.853)
Spline slope 4: 3.55 to 3.84				-13.21**
				(6.540)
Spline slope 5: above 3.84				-5.213
				(4.818)
Observations	1985	1985	1985	1985
R-squared	0.554	0.555	0.558	0.557
Mean of Dependent Variable	18.12	18.12	18.12	18.12
Full controls	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES

Standard errors clustered at district level in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

All columns include the full control set of Table 2 and State FE × Year FE. Spline knots at the quintile cutpoints of the lagged score; spline coefficients are within-segment slopes.

Table 23: Functional-form robustness: total number of private schools

	Total No. of Private Schools			
	Quadratic	Quintile bins	Decile bins	Spline
	(1)	(2)	(3)	(4)
Avg ASER learning in govt schools$_{t-1}$	854.1**			
	(428.3)			
Avg ASER learning in govt schools$^2_{t-1}$	-138.4**			
	(60.44)			
Score quintile (base: Q1, lowest)				
Q2		-8.563		
		(35.88)		
Q3		-49.49		
		(40.21)		
Q4		-87.00**		
		(39.78)		
Q5		-126.9***		
		(43.34)		
Score decile (base: D1, lowest)				
D2			89.76*	
			(54.38)	
D3			60.09	
			(54.18)	
D4			26.05	
			(75.08)	
D5			39.38	
			(61.05)	
D6			-31.18	
			(73.05)	
D7			-22.50	
			(63.77)	
D8			-50.75	
			(66.46)	
D9			-84.64	
			(68.58)	
D10			-65.75	
			(70.00)	
Spline slope 1: below 3.04				228.9*
				(132.1)
Spline slope 2: 3.04 to 3.31				-250.0*
				(139.9)
Spline slope 3: 3.31 to 3.55				-147.3
				(137.0)
Spline slope 4: 3.55 to 3.84				-150.9
				(118.3)
Spline slope 5: above 3.84				-70.97
				(86.75)
Observations	1985	1985	1985	1985
R-squared	0.756	0.755	0.757	0.757
Mean of Dependent Variable	523.8	523.8	523.8	523.8
Full controls	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES

Standard errors clustered at district level in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

All columns include the full control set of Table 2 and State FE × Year FE. Spline knots at the quintile cutpoints of the lagged score; spline coefficients are within-segment slopes.

Table 24: Functional-form robustness: private-school quality margins (score quintiles)

	Enrollment/school (z)	Grad-teacher share	Infra index
	(1)	(2)	(3)
Score quintile (base: Q1, lowest)			
Q2	-0.0467	-0.00987	-0.0429
	(0.0508)	(0.00897)	(0.0352)
Q3	0.0736	-0.00710	-0.0532
	(0.0760)	(0.0122)	(0.0453)
Q4	0.0628	0.00166	0.0303
	(0.0831)	(0.0128)	(0.0512)
Q5	0.0920	0.0179	0.0432
	(0.0986)	(0.0167)	(0.0591)
Observations	1985	1985	1985
R-squared	0.703	0.788	0.838
Mean of Dependent Variable	-0.00511	0.735	3.034
Full controls	YES	YES	YES
State FE × Year FE	YES	YES	YES

Standard errors clustered at district level in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

All columns include the full control set of Table 2 and State FE × Year FE.

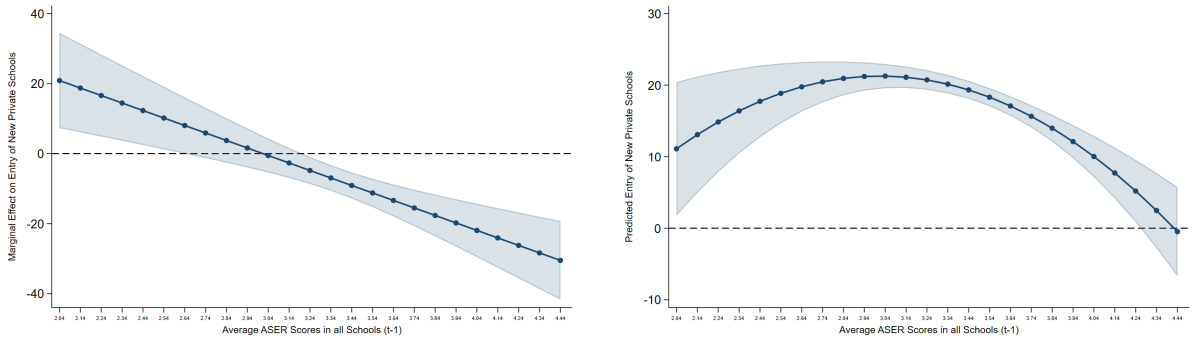


Figure 25: Lagged scores in all schools and entry of new private schools. 90 per cent confidence intervals.

Table 25: Heterogeneity (interactions): Private school entry. Full controls and state×year FE in all columns; SEs clustered by district.

	(1) Nightlights	(2) Parental educ.	(3) SC share	(4) ST share	(5) Rural share	(6) Grad+ tch/sch	Gra
Score_{t-1}	46.70**	53.86***	33.52**	45.19**	45.73**	67.05***	
	(20.10)	(19.56)	(16.83)	(21.09)	(20.91)	(21.76)	
Score^2_{t-1}	-7.918***	-9.173***	-5.721**	-8.554***	-7.869***	-11.31***	
	(2.930)	(2.912)	(2.473)	(3.043)	(2.969)	(3.290)	
Above median	-26.54	22.15	-40.66	-42.60	-15.75	54.53	
	(47.25)	(55.02)	(43.01)	(49.10)	(46.73)	(48.25)	
Above median × Score_{t-1}	16.98	-13.53	31.43	13.69	11.64	-34.82	
	(27.83)	(31.77)	(25.87)	(28.97)	(27.72)	(28.75)	
Above median × Score^2_{t-1}	-2.994	1.978	-5.458	-0.695	-1.846	5.505	
	(4.046)	(4.567)	(3.813)	(4.229)	(4.064)	(4.228)	
Observations	1985	1957	1985	1985	1985	1985	
R-squared	0.558	0.555	0.557	0.560	0.556	0.555	
Mean of Dep. Var.	18.12	18.27	18.12	18.12	18.12	18.12	
Joint p (interactions)	0.316	0.909	0.0593	0.00949	0.856	0.312	
Turning pt, below-median	2.949	2.936	2.930	2.642	2.905	2.965	
Turning pt, above-median	2.918	2.803	2.905	3.183	2.952	2.778	

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 26: Heterogeneity (interactions): Total no. of private schools. Full controls and state×year FE in all columns; SEs clustered by district.

	(1) Nightlights	(2) Parental educ.	(3) SC share	(4) ST share	(5) Rural share	(6) Grad+ tch/sch	Gra
Score_{t-1}	1104.8***	854.7*	351.9	817.9	-274.2	902.0	
	(426.5)	(478.8)	(430.1)	(586.3)	(472.5)	(557.4)	
Score^2_{t-1}	-170.4***	-137.2*	-58.81	-150.5*	16.34	-145.8*	
	(60.99)	(70.05)	(62.12)	(81.46)	(65.52)	(83.03)	
Above median	993.8	1963.8*	-1344.6	-603.5	-2966.4***	844.2	
	(987.5)	(1163.2)	(1047.3)	(1176.4)	(944.1)	(1142.9)	
Above median × Score_{t-1}	-511.0	-1047.2	936.7	111.5	1681.7***	-428.0	
	(579.9)	(661.8)	(627.9)	(663.6)	(555.6)	(677.7)	
Above median × Score^2_{t-1}	56.34	141.3	-155.4*	10.63	-234.1***	56.60	
	(84.11)	(93.87)	(92.31)	(93.32)	(80.96)	(99.31)	
Observations	1985	1957	1985	1985	1985	1985	
R-squared	0.760	0.758	0.758	0.761	0.759	0.757	
Mean of Dep. Var.	523.8	527.8	523.8	523.8	523.8	523.8	
Joint p (interactions)	0.0500	0.208	0.0440	0.0329	0.00574	0.719	
Turning pt, below-median	3.242	3.116	2.991	2.718	8.390	3.092	
Turning pt, above-median	2.604	23.02	3.008	3.323	3.231	2.656	

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 27: Heterogeneity (split samples at baseline median): Private school entry. Full controls and state×year FE; SEs clustered by district.

	(1) Nightlights LO	(2) Nightlights HI	(3) Parental educ. LO	(4) Parental educ. HI	(5) SC share LO	(6) SC share
Score_{t-1}	37.36*	72.39***	51.59***	46.31	26.49	83.71***
	(19.39)	(24.84)	(16.81)	(31.93)	(17.72)	(24.35)
Score^2_{t-1}	-6.003**	-12.26***	-9.289***	-7.835*	-4.814*	-13.73**
	(2.793)	(3.646)	(2.561)	(4.493)	(2.598)	(3.562)
N	956	1014	946	995	933	1037
R-squared	0.493	0.589	0.541	0.601	0.722	0.482
Turning point	3.112	2.952	2.777	2.955	2.752	3.048

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 28: Heterogeneity (split samples at baseline median): Total no. of private schools. Full controls and state×year FE; SEs clustered by district.

	(1) Nightlights LO	(2) Nightlights HI	(3) Parental educ. LO	(4) Parental educ. HI	(5) SC share LO	(6) SC share
Score_{t-1}	338.9	1023.4	459.3	42.64	-278.6	1592.7**
	(530.7)	(721.5)	(444.9)	(801.0)	(528.1)	(638.8)
Score^2_{t-1}	-52.69	-173.2*	-87.60	-19.29	26.54	-242.7**
	(75.58)	(103.2)	(66.34)	(110.2)	(75.87)	(91.43)
N	956	1014	946	995	933	1037
R-squared	0.739	0.780	0.757	0.802	0.762	0.763
Turning point	3.216	2.954	2.622	1.105	5.248	3.282

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 29: Robustness: ordinal ASER measures (proportions clearing ladder rungs, govt students) — Private school entry

	Score (baseline)	Subtraction+		Division		Reads Std I+		Reads Std II	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Avg ASER score, $govt_{t-1}$	49.76***								
	(16.26)								
Avg ASER score, $govt^2_{t-1}$	-8.523***								
	(2.377)								
Prop. clearing rung, $govt_{t-1}$		-19.69***	16.23	-24.49***	-8.754	-21.02***	61.44**	-18.76***	28.03
		(5.543)	(18.88)	(6.878)	(18.14)	(6.871)	(28.69)	(6.186)	(21.07)
Prop. clearing rung, $govt^2_{t-1}$			-41.42**		-28.89		-76.45***		-54.34**
			(19.85)		(27.80)		(24.40)		(21.71)
Full controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	1985	1985	1985	1985	1985	1985	1985	1985	1985
R-squared	0.554	0.550	0.551	0.550	0.550	0.550	0.554	0.550	0.552
Mean of Dep. Var.	18.12	18.12	18.12	18.12	18.12	18.12	18.12	18.12	18.12
Turning point	2.919		0.196		-0.151		0.402		0.258

All columns estimated on the same sample. Controls as in the headline spec: parental education, govt schools, govt enrollment, govt infra index, nightlights (all $t-1$); census infra + trend; SC/ST shares + trends; log pop + trend; blocks & villages. Standard errors clustered at district level in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$

Table 30: Robustness: ordinal ASER measures (proportions clearing ladder rungs, govt students) — Total no. of private schools

	Score (baseline)	Subtraction+		Division		Reads Std I+		Reads Std II	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Avg ASER score, $govt_{t-1}$	847.8**								
	(423.9)								
Avg ASER score, $govt^2_{t-1}$	-140.1**								
	(60.20)								
Prop. clearing rung, $govt_{t-1}$		-244.1**	251.0	-308.2**	-160.9	-249.7*	929.5	-255.9**	83.53
		(107.9)	(442.1)	(131.1)	(391.9)	(134.8)	(734.5)	(111.9)	(524.8)
Prop. clearing rung, $govt^2_{t-1}$			-571.1		-270.3		-1093.1*		-394.2
			(442.5)		(597.0)		(609.8)		(554.3)
Full controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	1985	1985	1985	1985	1985	1985	1985	1985	1985
R-squared	0.756	0.754	0.754	0.754	0.754	0.754	0.755	0.754	0.754
Mean of Dep. Var.	523.8	523.8	523.8	523.8	523.8	523.8	523.8	523.8	523.8
Turning point	3.025		0.220		-0.298		0.425		0.106

All columns estimated on the same sample. Controls as in the headline spec: parental education, govt schools, govt enrollment, govt infra index, nightlights (all $t-1$); census infra + trend; SC/ST shares + trends; log pop + trend; blocks & villages. Standard errors clustered at district level in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$

Table 31: Quintile-bin (rank-only) version: within-year quintiles of lagged govt-school learning — Private school entry

	Score	Subtraction+	Division	Reads Std I+	Reads Std II
	(1)	(2)	(3)	(4)	(5)
Quintile 2 of measure$_{t-1}$	-0.909	1.072	0.0483	0.176	1.420
	(2.070)	(1.672)	(1.647)	(1.943)	(1.881)
Quintile 3 of measure$_{t-1}$	-3.514	-1.673	-0.667	-3.001	-1.172
	(2.173)	(2.072)	(2.015)	(2.190)	(2.036)
Quintile 4 of measure$_{t-1}$	-8.283***	-5.431***	-5.625***	-7.129***	-3.797
	(2.287)	(2.059)	(2.017)	(2.152)	(2.317)
Quintile 5 (highest) of measure$_{t-1}$	-10.36***	-7.329***	-6.463***	-10.07***	-7.775***
	(2.606)	(2.283)	(2.352)	(2.494)	(2.390)
Full controls	YES	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES	YES
N	1985	1985	1985	1985	1985
R-squared	0.555	0.552	0.552	0.555	0.553
Mean of Dep. Var.	18.12	18.12	18.12	18.12	18.12

Omitted category: bottom quintile. Quintiles computed within survey year, so estimates use only the ordering of districts, not the cardinal scale. Controls as in the headline spec. Standard errors clustered at district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 32: Quintile-bin (rank-only) version: within-year quintiles of lagged govt-school learning — Total no. of private schools

	Score	Subtraction+	Division	Reads Std I+	Reads Std II
	(1)	(2)	(3)	(4)	(5)
Quintile 2 of measure$_{t-1}$	-15.21	1.939	16.91	5.455	1.559
	(37.52)	(31.81)	(30.20)	(42.05)	(35.51)
Quintile 3 of measure$_{t-1}$	-61.06	-26.25	-14.19	-75.25	-41.96
	(39.04)	(40.52)	(36.34)	(46.37)	(44.34)
Quintile 4 of measure$_{t-1}$	-97.07**	-53.99	-45.71	-101.7**	-63.87
	(42.93)	(43.69)	(41.70)	(45.14)	(44.85)
Quintile 5 (highest) of measure$_{t-1}$	-137.2***	-94.61**	-47.95	-128.2***	-118.2***
	(44.32)	(42.84)	(52.23)	(45.68)	(44.11)
Full controls	YES	YES	YES	YES	YES
State FE × Year FE	YES	YES	YES	YES	YES
N	1985	1985	1985	1985	1985
R-squared	0.755	0.754	0.754	0.756	0.755
Mean of Dep. Var.	523.8	523.8	523.8	523.8	523.8

Omitted category: bottom quintile. Quintiles computed within survey year, so estimates use only the ordering of districts, not the cardinal scale. Controls as in the headline spec. Standard errors clustered at district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 33: Effect of lagged ASER learning levels in all schools on total no. of private schools

	Total Private Schools		
	(1)	(2)	(3)
Avg ASER learning in all schools$_{t-1}$	-166.6***	-6.346	1282.6***
	(54.07)	(51.43)	(456.7)
Avg ASER learning in public schools$^2_{t-1}$			-194.0***
			(65.13)
Average Parental Education$_{t-1}$		29.23	25.34
		(18.51)	(18.36)
Total no. of public schools$_{t-1}$		0.0391	0.0472
		(0.0529)	(0.0519)
Normalized enrollment in public schools$_{t-1}$		-70.80	-71.06
		(43.58)	(43.19)
Infrastructure index in public schools$_{t-1}$		-16.20	-14.67
		(48.76)	(48.42)
Light Intensity$_{t-1}$		239.1***	236.6***
		(39.99)	(40.39)
Census infra index		-98.23**	-95.13**
		(41.01)	(40.46)
Census infra index × Trend		-2.371	-3.595
		(3.664)	(3.585)
Log(Population)		319.4***	321.4***
		(67.37)	(66.16)
Log(Population) x Trend		-29.65***	-30.44***
		(10.47)	(10.36)
% of SC population		159.4	98.54
		(306.8)	(299.5)
% of SC population × Trend		30.04	28.84
		(36.35)	(36.09)
% of ST population		11.98	-18.35
		(127.3)	(123.5)
% of ST population × Trend		-3.461	-0.0546
		(12.60)	(12.42)
Constant	1091.3***	-964.5**	-3013.6***
	(189.9)	(453.5)	(929.1)
Observations	2514	1985	1985
R-squared	0.0112	0.753	0.755
Mean of Dependent Variable	519.0	523.8	523.8
# of Blocks & Villages	NO	YES	YES
Year FE	NO	YES	YES
State FE	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 34: Effect of lagged average ASER learning levels in all schools on the entry of private schools

	Private School Entry		
	(1)	(2)	(3)
Avg ASER learning$_{t-1}$	-10.09***	-6.592***	64.51***
	(1.992)	(2.299)	(20.33)
Avg ASER learning$^2_{t-1}$			-10.70***
			(3.005)
Average Parental Education$_{t-1}$		1.321	1.107
		(0.940)	(0.916)
Total no. of govt schools$_{t-1}$		0.00606**	0.00650***
		(0.00236)	(0.00232)
Normalized enrollment in govt schools$_{t-1}$		4.145**	4.130**
		(1.969)	(1.951)
Infrastructure index in govt schools$_{t-1}$		-2.340	-2.255
		(1.962)	(1.939)
Light Intensity$_{t-1}$		8.008***	7.867***
		(1.409)	(1.461)
Census infra index		-2.681	-2.510
		(2.777)	(2.755)
Census infra index × Trend		0.151	0.0837
		(0.416)	(0.414)
log(Rural population)		10.92***	11.03***
		(3.248)	(3.211)
log(Rural population) × Trend		-1.688***	-1.732***
		(0.565)	(0.565)
% of SC population		-0.579	-3.937
		(22.06)	(21.82)
% of SC population × Trend		3.194	3.127
		(3.472)	(3.456)
% of ST population		-14.87*	-16.55*
		(8.807)	(8.684)
% of ST population × Trend		1.004	1.191
		(1.348)	(1.343)
Constant	52.26***	17.12	-95.92**
	(6.985)	(21.76)	(44.25)
Observations	2019	1985	1985
R-squared	0.0230	0.550	0.554
Mean of Dependent Variable	17.88	18.12	18.12
# of Blocks & Villages	NO	YES	YES
Year FE	NO	YES	YES
State FE	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 35: Effect of lagged ASER learning levels in public schools on ratio of enrolment in new private schools to enrolment in public schools

	New Private School Enrollment/Public School Enrollment		
	(1)	(2)	(3)
Avg ASER learning in public schools$_{t-1}$	0.0349	0.149***	-0.0511
	(0.0601)	(0.0521)	(0.375)
Avg ASER learning in public schools$^2_{t-1}$			0.0302
			(0.0582)
Average Parental Education$_{t-1}$		0.0792***	0.0797***
		(0.0211)	(0.0211)
Total no. of public schools$_{t-1}$		-0.000275***	-0.000276***
		(0.0000535)	(0.0000533)
Infrastructure index in public schools$_{t-1}$		0.114**	0.114**
		(0.0474)	(0.0474)
Light Intensity$_{t-1}$		0.280***	0.281***
		(0.0267)	(0.0268)
Census infra index		-0.0982*	-0.0976*
		(0.0563)	(0.0564)
Census infra index × Trend		0.00888	0.00901
		(0.00666)	(0.00661)
Log(Population)		0.383***	0.383***
		(0.0752)	(0.0754)
Log(Population) x Trend		-0.00190	-0.00174
		(0.00616)	(0.00618)
% of SC population		-0.0810	-0.0689
		(0.512)	(0.518)
% of SC population × Trend		-0.0203	-0.0201
		(0.0677)	(0.0675)
% of ST population		0.0173	0.0207
		(0.205)	(0.207)
% of ST population × Trend		0.0200	0.0199
		(0.0268)	(0.0269)
Constant	0.656***	-2.976***	-2.668***
	(0.227)	(0.503)	(0.802)
Observations	2514	1985	1985
R-squared	0.0211	0.736	0.736
Mean of Dependent Variable	0.628	0.651	0.651
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$